18. A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ωt, where t represents time elapsed : CBSE 2026

(a) Magnetic flux ϕ linked with the coil, and

(b) emf induced in the coil. *2*

18. A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ωt, where t represents time elapsed :

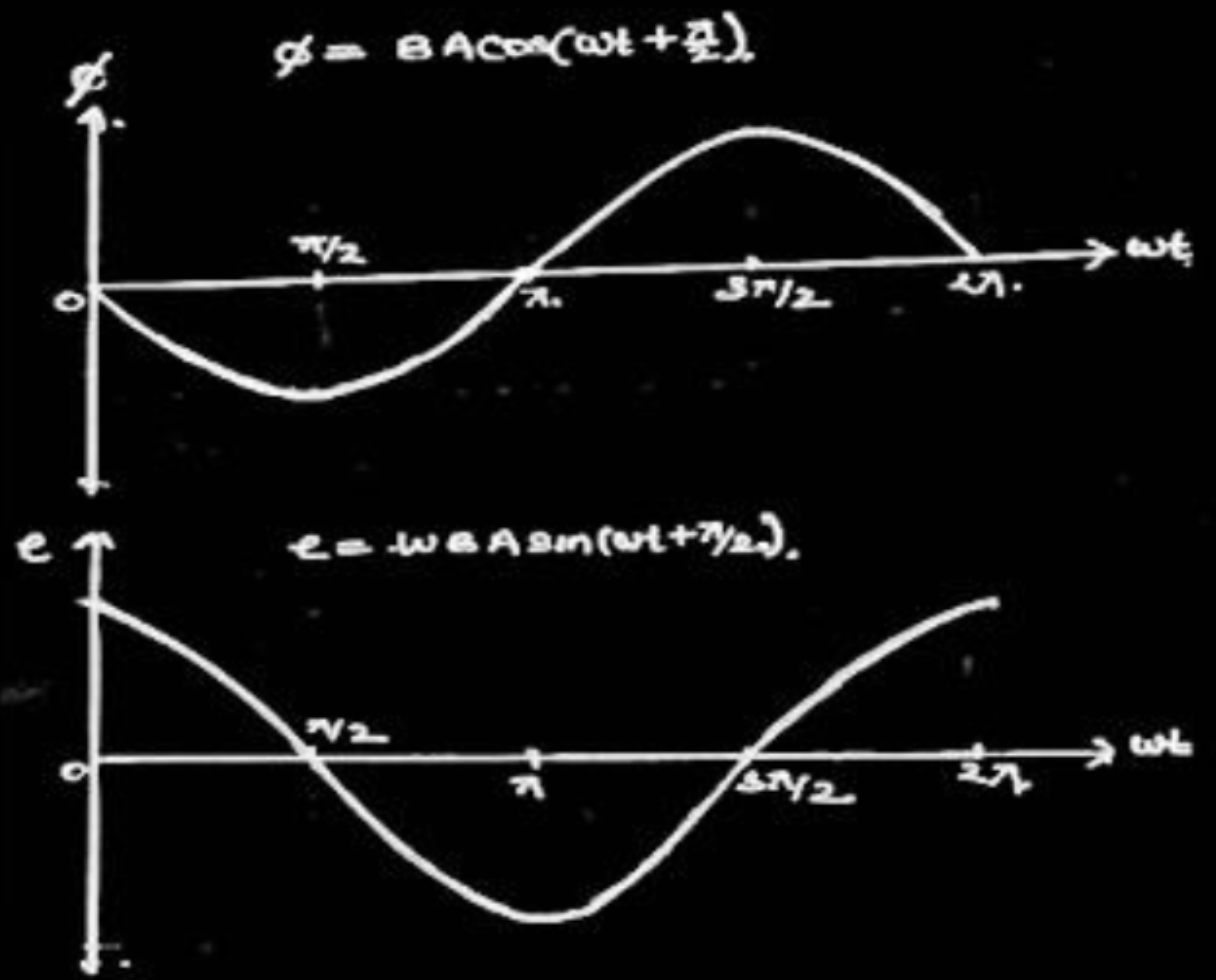
CBSE 2026

(a) Magnetic flux ϕ linked with the coil, and

(b) emf induced in the coil. *2*

(a)

(b)



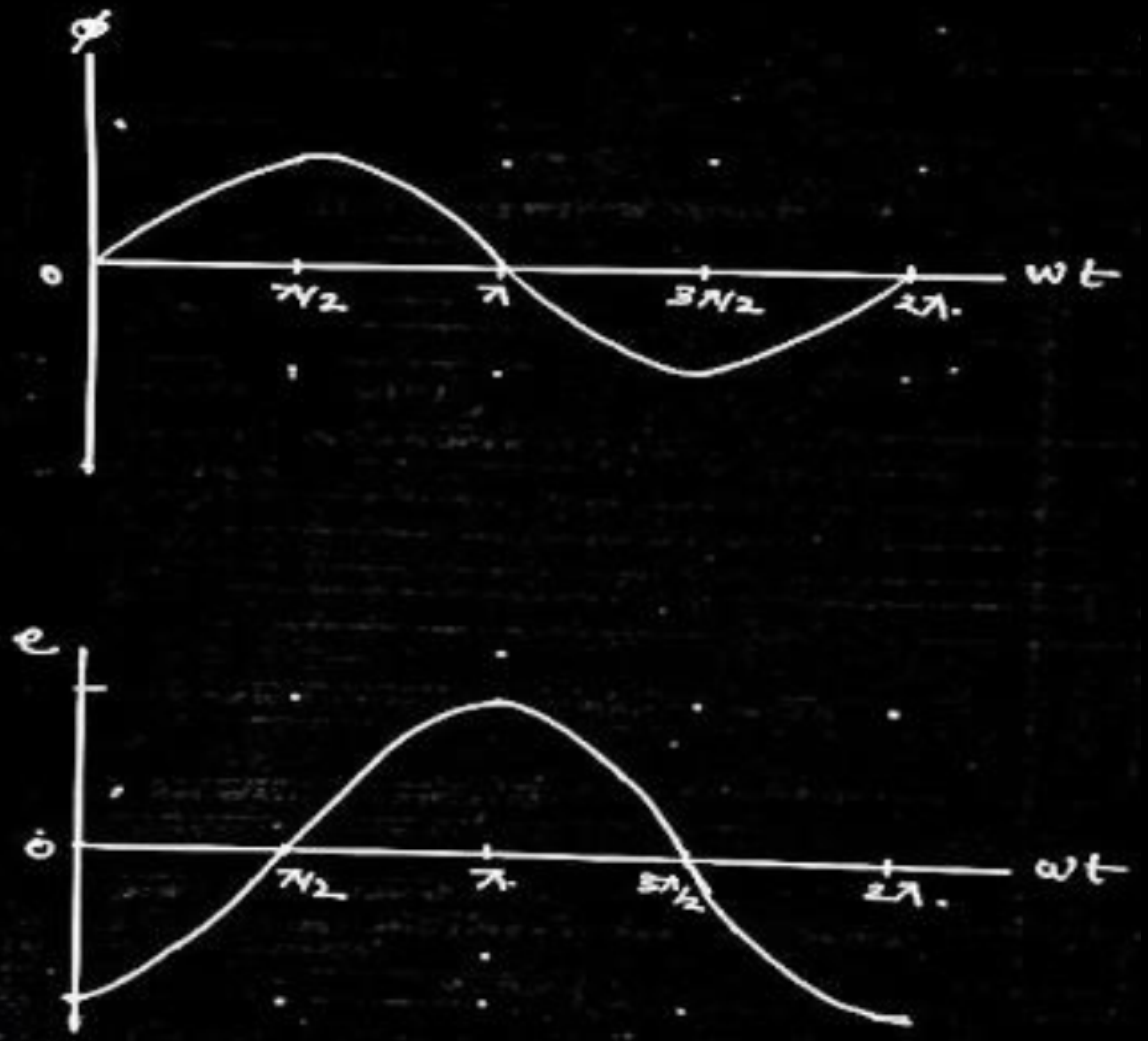
1

1

Alternatively,

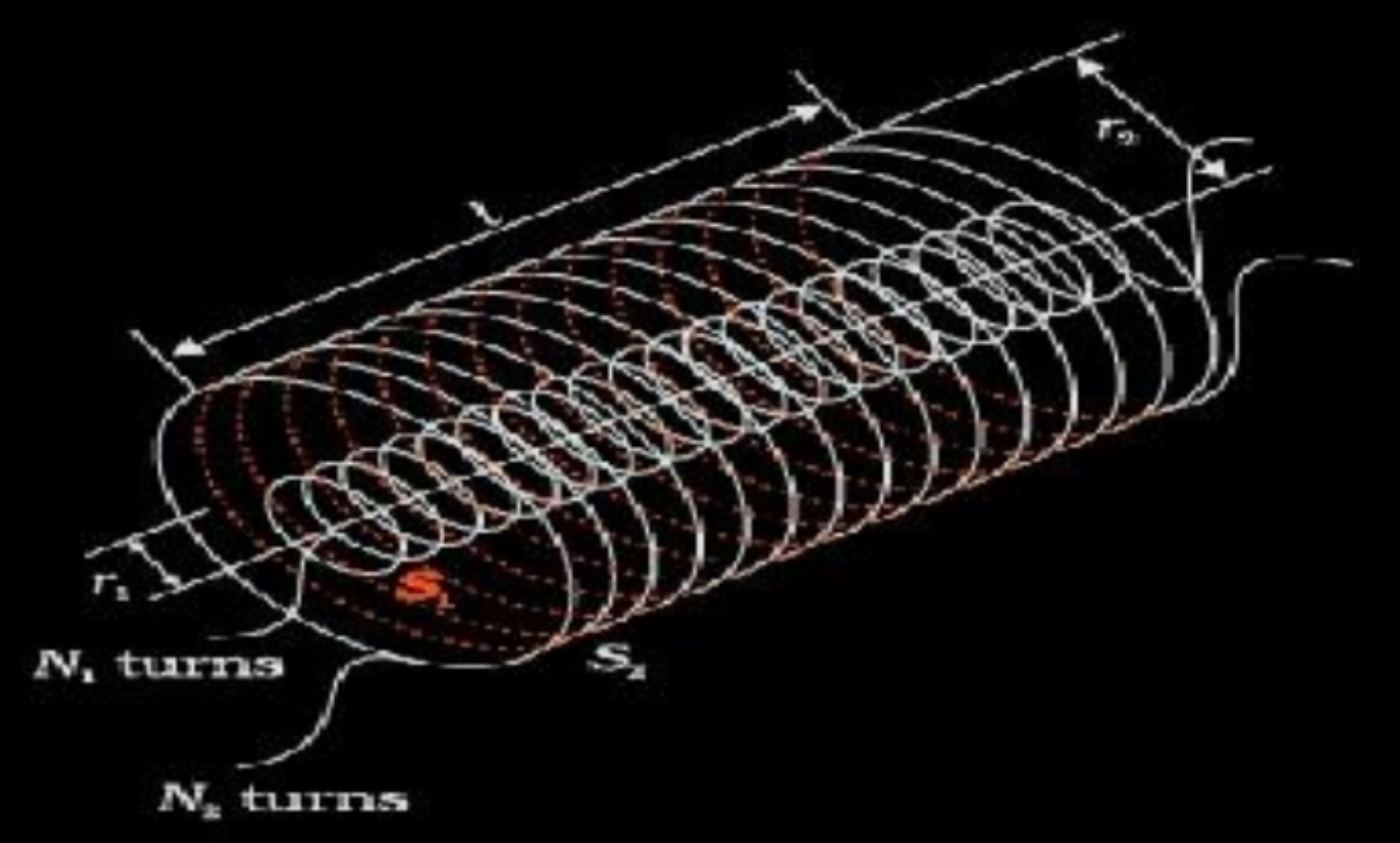
(a)

(b)



24. A long solenoid of length L and radius r_1 having N_1 turns is surrounded symmetrically by a coil of radius r_2 ($> r_1$) having N_2 turns ($N_2 << N_1$) around its mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is $M_{12} = M_{21}$ valid in this case ? CBSE 2026 *3*

24. A long solenoid of length L and radius r_1 having N_1 turns is surrounded symmetrically by a coil of radius r_2 $(> r_1)$ having N_2 turns $(N_2 << N_1)$ around its mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is $M_{12} = M_{21}$ valid in this case ? CBSE 2026 *3*



Let n_1 and n_2 be the total no. of turns per unit length of solenoid S_1 and coil S_2 respectively

$N_1 = n_1 L, \qquad N_2 = n_2 L$

When current I_2 is set up in S_2, flux linkage with S_1 is:

$N_1\phi_1 = N_1 A_1 B_2$ ....(i) ½

$N_1\phi_1 = (n_1 L)(\pi r_1^2)(\mu_o n_2 I_2)$

$N_1\phi_1 = (\mu_o n_1 n_2 \pi r_1^2 L I_2)$ ....(ii) ½

$N_1\phi_1 = M_{12} I_2$ ....(iii) ½

From (ii) and (iii)

$M_{12} = (\mu_o n_1 n_2 \pi r_1^2 L)$ ½

Considering the reverse case, when I_1 current is set up in S_1, flux linkage with S_2 is

$N_2\phi_2 = N_2 A_1 B_1$ .... (iv)

$N_2\phi_2 = (n_2 L)(\pi r_1^2)(\mu_o n_1 I_1)$ ....(v)

$N_2\phi_2 = M_{21} I_1$ ....(vi)

From (v) and (vi)

$M_{21} = (\mu_o n_1 n_2 \pi r_1^2 L)$ ½

Therefore, $M_{12} = M_{21}$ ½

Alternatively,

$M_{12} = M_{21} = \dfrac{(\mu_o N_1 N_2 \pi r_1^2)}{L}$

20. A square loop of side 10 cm, free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of 0·2 T. If it is rotated at the uniform speed of 60 rpm, find the emf induced in the loop. CBSE 2026 *2*

20. A square loop of side 10 cm, free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of 0·2 T. If it is rotated at the uniform speed of 60 rpm, find the emf induced in the loop. CBSE 2026 *2*

$e = BA\omega \sin \omega t$	½
$e = 0.2 \times 100 \times 10^{-4} \times 2\pi \times \sin 2\pi t$	1
$e = 4\pi \times 10^{-3} \sin 2\pi t$ V	½
$= 12.56 \times 10^{-3} \sin 2\pi t$ V	
Note: Award full marks if $\sin 2\pi t$ **is not mentioned in the final answer.**	

(b) A rectangular loop of sides l and b and resistance 'R' is kept in a region in which the magnetic field varies as $B = B_0 \sin \omega t$.

CBSE 2026

(i) Derive expression for the emf induced in the loop.

(ii) Find the effective value of current that flows in the loop. 3

(b) A rectangular loop of sides l and b and resistance 'R' is kept in a region in which the magnetic field varies as $B = B_0 \sin \omega t$.

CBSE 2026

(i) Derive expression for the emf induced in the loop.

(ii) Find the effective value of current that flows in the loop. *3*

(i) Deriving expression for induced emf	1 ½
(ii) Finding the effective value of current	1 ½

(i) Induced emf $= |\varepsilon| = \dfrac{d\phi}{dt}$ ½

$$= \frac{d}{dt}\left(\vec{B}.\vec{A}\right)$$

$$|\varepsilon| = \frac{d}{dt}\left(B_0 \sin \omega t \times \ell b\right)$$ ½

$$= B_0 \omega \ell b \cos \omega t$$ ½

(ii) Instantaneous value of current $= \dfrac{|\varepsilon|}{R} = \dfrac{B_0 \omega \ell b \cos \omega t}{R}$ ½

$$I_{eff} = \frac{I_0}{\sqrt{2}}$$ ½

$$= \frac{B_0 \omega \ell b}{R\sqrt{2}}$$ ½

2. A closely wound long solenoid of self-inductance L is cut into two identical solenoids. The value of self-inductance of each small solenoid will be :

(A) $\frac{L}{2}$ (B) 2L

(C) 3L (D) 4L

CBSE 2026

2. A closely wound long solenoid of self-inductance L is cut into two identical solenoids. The value of self-inductance of each small solenoid will be :

(A) $\frac{L}{2}$

(B) 2L

CBSE 2026

(C) 3L

(D) 4L

(A) $\frac{L}{2}$

9. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the :

(A) rate at which currents change in the two coils.

(B) relative position and orientation of the coils.

(C) currents in the two coils.

(D) value of voltage induced in one coil due to change in value of current in the other coil.

CBSE 2026

9. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the :

(A) rate at which currents change in the two coils.

(B) relative position and orientation of the coils.

(C) currents in the two coils.

(D) value of voltage induced in one coil due to change in value of current in the other coil.

CBSE 2026

(B) relative position and orientation of the coils

28. Two coils, one of radius 0·5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate :

(a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and

(b) the mutual inductance of the two coils. *3*

CBSE 2026

28. Two coils, one of radius 0·5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate :

(a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and

(b) the mutual inductance of the two coils. *3*

CBSE 2026

(a) $\phi_1 = N_1 B_2 A_1$	½
$= N_1 \times \dfrac{\mu_0 I_2}{2r_2} \times \pi r_1^2 \times N_2$	½
$= \dfrac{10\times4\pi\times10^{-7}\times3\times3.14\times0.5\times0.5\times10^{-4}\times50}{2\times5\times10^{-2}}$	½
$=14.789\times10^{-7}$ Wb	
(b) $M_{12} = \dfrac{\phi_1}{I_2}$	½
$= \dfrac{14.789\times10^{-7}}{3}$	½
$=4.92\times10^{-7}$ H	½

(b) (i) State Lenz's law and explain that it follows the law of conservation of energy.

(ii) Write the dimensional formula for self-inductance.

The current in a coil changes from 8·0 A to 2·0 A in 0·6 s. If an average emf induced in the coil is 50 V, calculate the self-inductance of the coil. 5

CBSE 2026

(i)	The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it. Consider north pole of a magnet being moved towards a coil. The magnetic flux through the coil increases, hence current is induced in counter clockwise direction making the side of the coil behave as north pole causing repulsion. If this is not the case then bringing magnet towards a coil will construct a perpetual motion machine by a suitable arrangement. Therefore production of induced emf in a direction which opposes the change in magnetic flux follows from law of conservation of energy.	1 1
(ii)	$[\,ML^2T^{-2}A^{-2}\,]$ $\varepsilon = -L\dfrac{dI}{dt}$ $L = \dfrac{-(50)\times 0.6}{(2-8)}$ $= 5\ H$	1 ½ 1 ½

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.

(b) Considering the case of the magnetic field B inside a solenoid of length l, area of cross-section A, prove that magnetic energy per unit volume is $B^2/(2\mu_o)$. CBSE 2026 **3**

24. (a) Briefly explain how Lenz's law is consistent with the law of conservation of energy.

(b) Considering the case of the magnetic field B inside a solenoid of length l, area of cross-section A, prove that magnetic energy per unit volume is $B^2/(2\,\mu_o)$. **3**

CBSE 2026

(a) Lenz's law is consistent with the law of conservation of energy because: When a magnet is moved towards or away from a coil, then induced current creates its own magnetic field that opposes the motion of the magnet. Due to this opposition an external force must be applied to keep the magnet moving. The work done by this external force is converted into electrical energy.	1
(b) Magnetic field inside a solenoid, $B=\mu_0 nI$	½
Coefficient of self-inductance, $L=\mu_0 n^2 Al$	½
Magnetic energy $U_B=\frac{1}{2}LI^2$ $=\frac{1}{2}(\mu_0 n^2 Al)\left(\frac{B}{\mu_o n}\right)^2$ $=\frac{1}{2\mu_o}B^2 Al$	½
Magnetic energy per unit volume, $\frac{U_B}{Volume}=\frac{U_B}{Al}=\frac{B^2}{2\mu_0}$	½

32. (a) Define mutual inductance. Give its units. State two factors on which the mutual inductance between a given pair of coils depends.

(b) A 1.0 m long solenoid of radius 2 cm having 2000 turns has a secondary coil of 1000 turns wound closely near its mid-point. Find the mutual inductance between the two coils. **5**

CBSE 2026

(i) Mutual inductance is defined as the induced emf in primary coil when the current in secondary coil changes at the unit rate. Alternatively, Mutual inductance is defined as the magnetic flux linked with the primary coil when the current in secondary coil is unity.	1
S.I unit: henry (H) Alternatively, weber/ampere Alternatively, Tm^2A^{-1}	1
Factors: (1) Separation between two coils	½
(2) Relative orientation between two coils	½
(3) Nature of the medium (Any two)	
(ii) Mutual Inductance, $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{l}$	1
$= \dfrac{(4\pi\times10^{-7})(2000)(1000)(3.14)\left(2\times10^{-2}\right)^2}{1}$	½
$= 3.16\times\ 10^{-3}H$	½

25. (i) Define mutual inductance of a pair of coils. Write its SI unit. **3**

(ii) A long solenoid of radius R and length L has n turns per unit length. A circular loop of radius $r(<R)$ is placed inside at the centre of the solenoid such that its axis coincides with the axis of the solenoid. Obtain the mutual inductance of the solenoid and the loop.

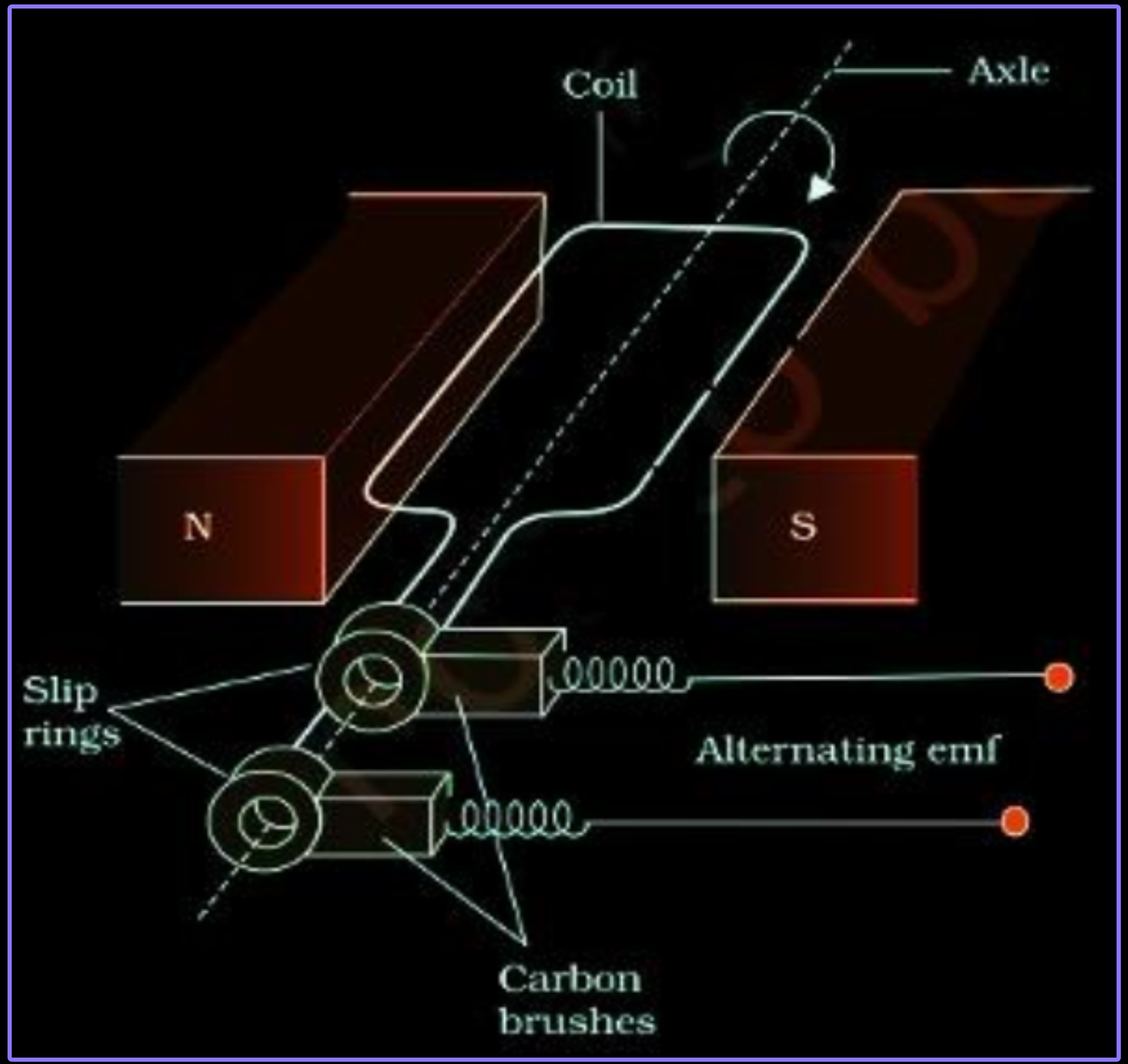
CBSE 2026

(i) It is the ratio of the magnetic flux linked with a coil to the current flowing in the neighbouring coil. **<u>Alternatively:</u>** It is the ratio of magnitude of induced emf in a coil to the rate of change of current in the neighbouring coil. **Note: Award (1/2) mark if a student writes** $M = \frac{\phi}{I}$ OR $M = \frac{\lvert\varepsilon\rvert}{di/dt}$ SI unit: henry(H) Or Tm^2A^{-1} Or WbA^{-1} or VsA^{-1}	1 1/2
(ii) If current I is setup through the solenoid, the magnetic field inside at its center $B = \mu_0 nI$ Flux linked with the circular loop of radius $r(< R)$ $\phi_B = BA$ $\phi_B = (\mu_0 nI)(\pi r^2)$ Mutual Inductance $M = \frac{\phi_B}{I}$ $M = \mu_0 n\pi r^2$	 1/2 1/2 1/2

(i) With the help of a labelled diagram, explain the principle, construction and working of an a.c. generator.

(ii) Deduce an expression for the induced emf in the coil of the generator.

(iii) If T is the time period of the rotation of the coil, at what values of T in a cycle, the emf generator is maximum ? CBSE 2026



- Principle: It is based on the principle of electromagnetic induction. 1

When the magnetic flux passing through a coil changes continuously, emf is induced.

Working: When coil is rotated in magnetic field, magnetic flux linked with it changes and emf is induced in the coil. 1

(ii) Flux linked with a coil at any instant of time placed in magnetic field rotating with angular speed ω

$$\phi_B = BA\cos\omega t$$ 1/2

From faraday's law

$$\varepsilon = -N\frac{d\phi_B}{dt}$$ 1/2

$$\varepsilon = NBA\omega \sin\omega t$$

(iii) EMF generated is maximum at $\frac{T}{4}$ and $\frac{3T}{4}$ 1

33. (a) State Faraday's law of electromagnetic induction. **5**

(b) Derive an expression for the self-inductance of an air-filled long solenoid of length l and cross-sectional area A having N turns.

(c) A conducting rod of length 50 cm, with one end pivoted, is rotated with angular speed of 60 rpm in a uniform magnetic field of 4.0 mT directed perpendicular to the plane of rotation of rod. Find the emf induced in the rod.

(a) The magnitude of the induced e.m.f in a circuit is equal to the time rate of change of magnetic flux through the circuit.	1
Alternatively:	
$\varepsilon = -\dfrac{d\phi_B}{dt}$	
(b) Magnetic field due to current carrying long solenoid of length l area of cross section A having n turns per unit length is $B = \mu_0 nI$	
Total magnetic flux linked with the solenoid is	½
$N\phi_B = (nl)(\mu_0 nI)(A)$ $= \mu_0 n^2 AlI$	½
Where nl is total number of turns. Thus	
$\therefore$ Self inductance of the solenoid	
$L = \dfrac{N\phi_B}{I}$	½
$L = \mu_0 n^2 Al$	½
Note: Award full marks for any other correct alternative method	
(c) Induced e.m.f	
$\varepsilon = \dfrac{1}{2} Bl^2\omega$	½
$= \dfrac{1}{2} \times 4 \times 10^{-3} \times \left(50 \times 10^{-2}\right)^2 \times (2\pi \times 1)$	1
$= 3.14\ \text{mV}$	½

OR

4. The dimensions of ‘self-inductance’ are :

(A) $[M\,L\,T^{-2}\,A^{-2}]$

(B) $[M\,L^{2}\,T^{-1}\,A^{-1}]$

(C) $[M\,L^{-1}\,T^{-2}\,A^{-2}]$

(D) $[M\,L^{2}\,T^{-2}\,A^{-2}]$

4. The dimensions of 'self-inductance' are :

(A) $[M\,L\,T^{-2}\,A^{-2}]$

(B) $[M\,L^{2}\,T^{-1}\,A^{-1}]$

(C) $[M\,L^{-1}\,T^{-2}\,A^{-2}]$

(D) $[M\,L^{2}\,T^{-2}\,A^{-2}]$

(D) $\left[ML^{2}T^{-2}A^{-2}\right]$

(b) Define 'self-inductance' of a coil. Derive an expression for self-inductance of a long solenoid of cross-sectional area A and length l, having n turns per unit length. *3*

Self inductance of a coil is the ratio of the flux linkage to the current flowing in the coil. **Alternatively:** Self inductance of a coil is defined as the flux linked with the coil when unit current flows through it. **Alternatively**: Self inductance of a coil may be defined as the magnitude of emf induced in the coil when current changes at the rate of 1 A/s in the coil.	1
Expression for self inductance of a long solenoid: The magnetic field due to current flowing in the solenoid, B = $\mu_0 n I$	½
Total flux linked with the given solenoid $N\phi_B = (nl)(\mu_0 n I)\,\mathrm{A}$ $N\phi_B = \mu_0 n^2\,\mathrm{A}\,l I$	½
Self inductance $L = \dfrac{N\phi_B}{I}$	½
$L = \mu_0 n^2\,\mathrm{A}\,l$	½

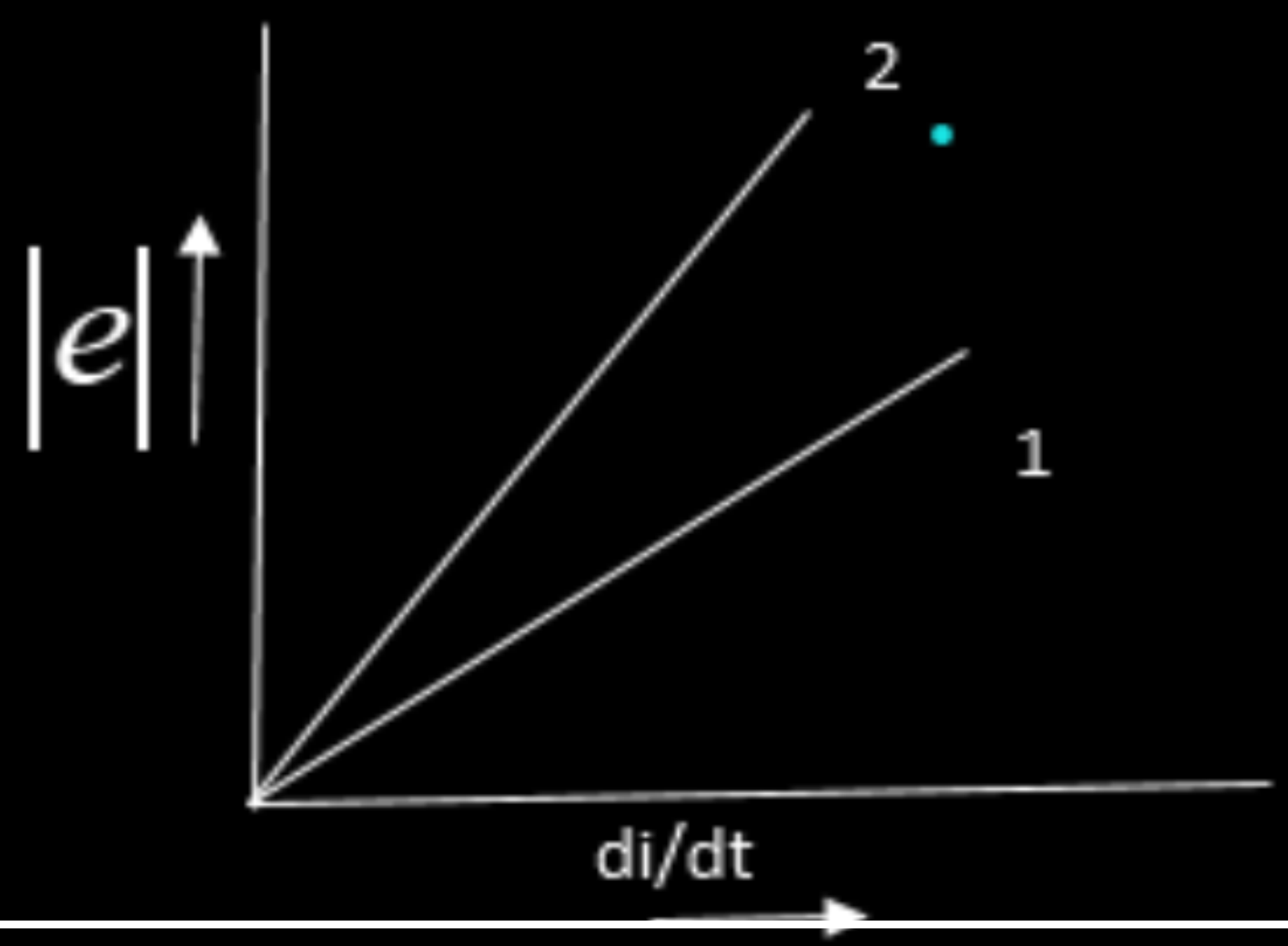
33. (a) (i) Define self-inductance of a coil. Derive the expression for the energy required to build up a current I in a coil of self-inductance L.

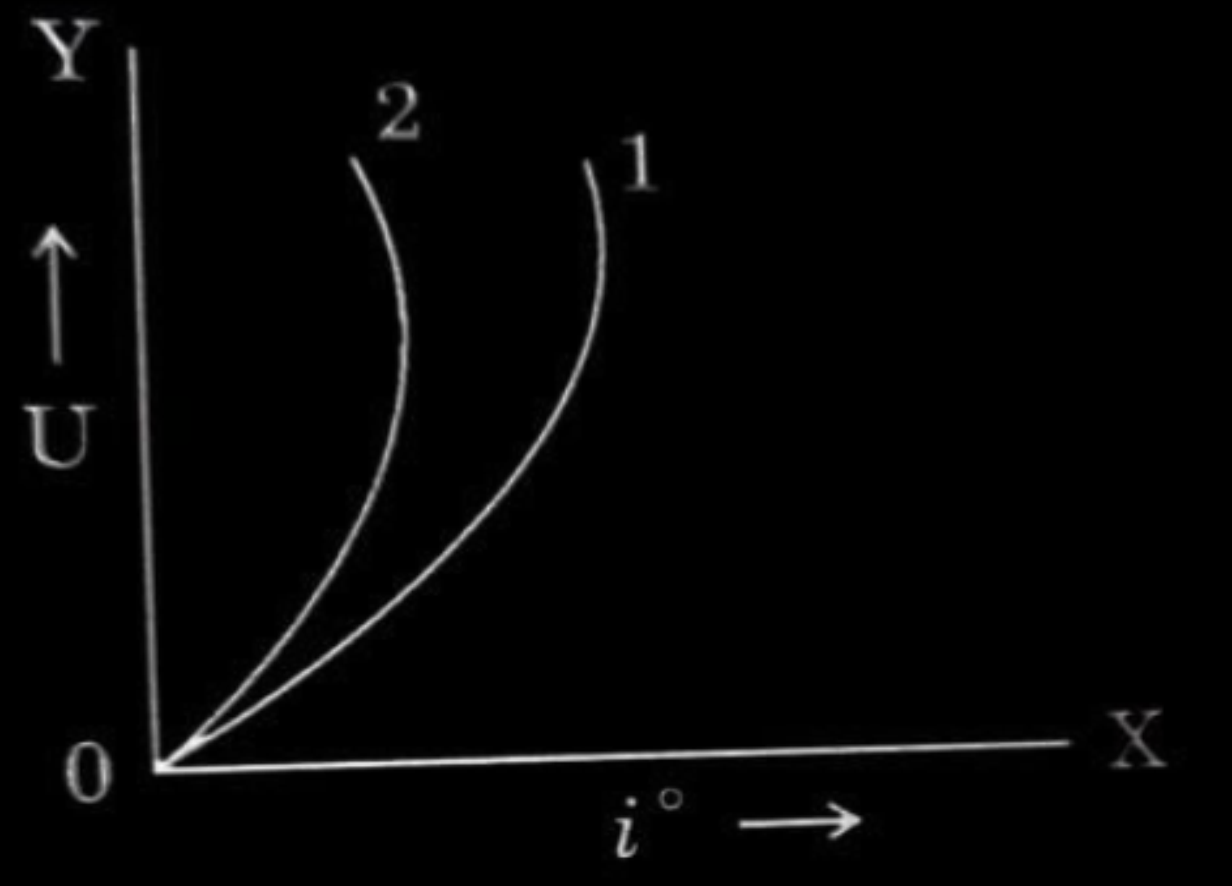
(ii) The currents passing through two inductors of self-inductances 10 mH and 20 mH increase with time at the same rate.

Draw graphs showing the variation of :

(I) the magnitude of emf induced with the rate of change of current in each inductor.

(II) the energy stored in each inductor with the current flowing through it. *5*

Self Inductance is magnetic flux linked with a coil when the current through the coil is unity. **Alternatively** Self Inductance is the induced emf induced in the coil when rate of change of current through the coil is unity. To maintain growth of current, power has to be supplied from external source.	1
$P = \lvert e \rvert \lvert I \rvert$ $= \frac{dW}{dt} = LI\frac{dI}{dt}$ $dW = LIdI$ $W = \int LIdI$	½
$= \frac{1}{2}LI^2$	½
(I) $E = -L\frac{dI}{dt}$	½
	1

(II) $U = \frac{1}{2}LI^2$ Parabolic graph obtained.	½
	1
(1 indicates 10mH) & (2 indicates 20mH)	

(b) (i) Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.

(ii) The current through an inductor is uniformly increased from zero to 2 A in 40 s. An emf of 5 mV is induced during this period. Find the flux linked with the inductor at t = 10 s. *5*

(b) (i) Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.

(ii) The current through an inductor is uniformly increased from zero to 2 A in 40 s. An emf of 5 mV is induced during this period. Find the flux linked with the inductor at t = 10 s. *5*

(i) Mutual inductance is defined as the induced emf in primary coil when the current in secondary coil changes at the unit rate. Alternatively Mutual inductance is defined as the magnetic flux linked with the primary coil when the current in secondary coil is unity.	**1**

Consider two long co-axial solenoids each of length l. Radius of inner solenoid S_1 is r_1 and number of turns per unit length is n_1.

The corresponding quantities for outer solenoid S_2 are r_2 and n_2 respectively. Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 respectively.

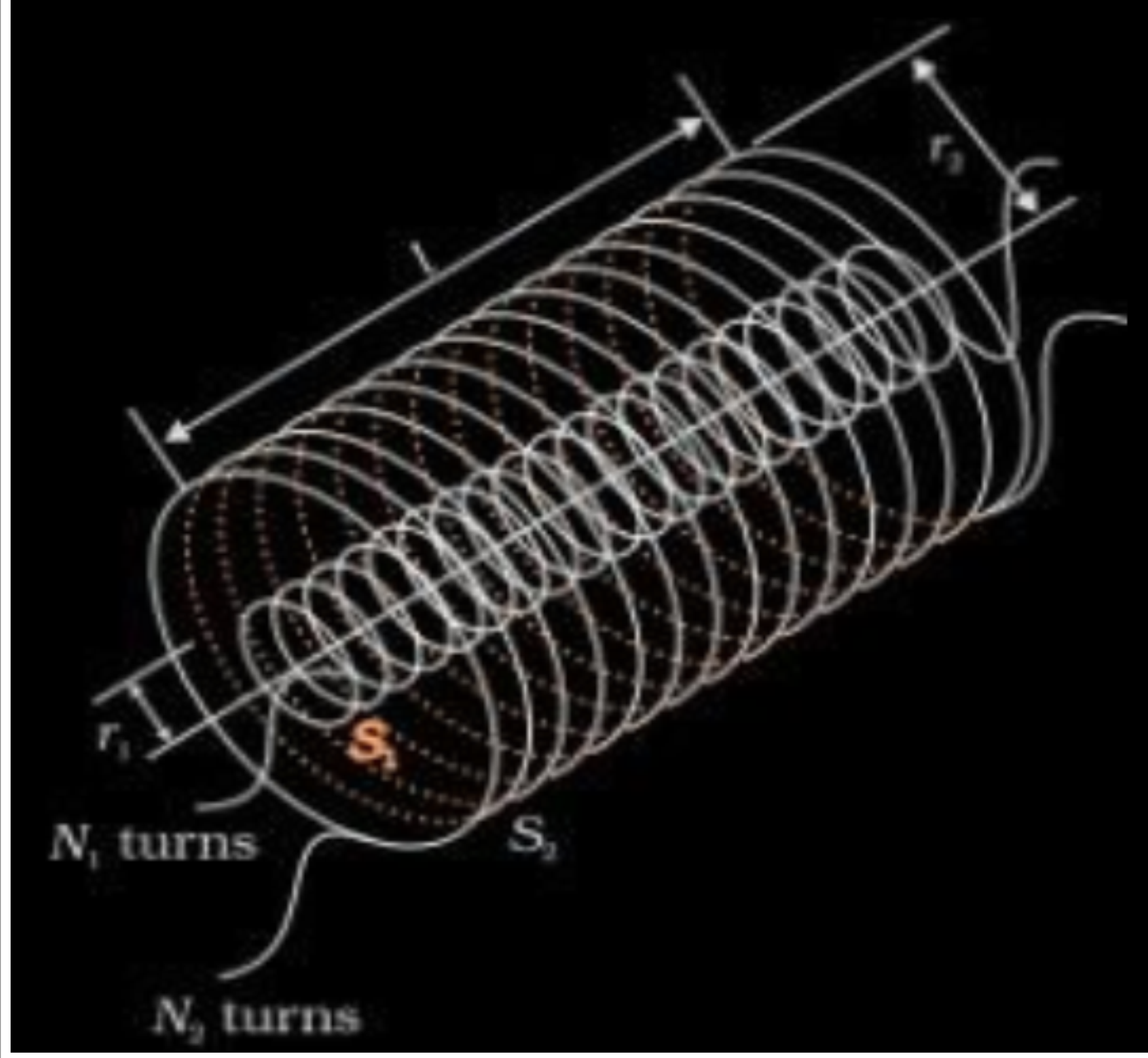
When a current I_2 is set up through S_2, it sets up magnetic flux through S_1.

$$N_1\phi_1 = M_{12}I_2$$ ½

$$= (n_1 l)\times(\pi r_1^2)\times(\mu_0 n_2 I_2)$$ ½

$$= \mu_0 n_1 n_2 \pi r_1^2 l I_2$$ ½

$$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l = M_{21}$$



½

(ii)

½

$$|e| = L\frac{dI}{dt}$$

$$L = \frac{e}{dI/dt}$$

$$= \frac{5\times10^{-3}}{2/40}$$

$$= 0.1\text{H}$$ ½

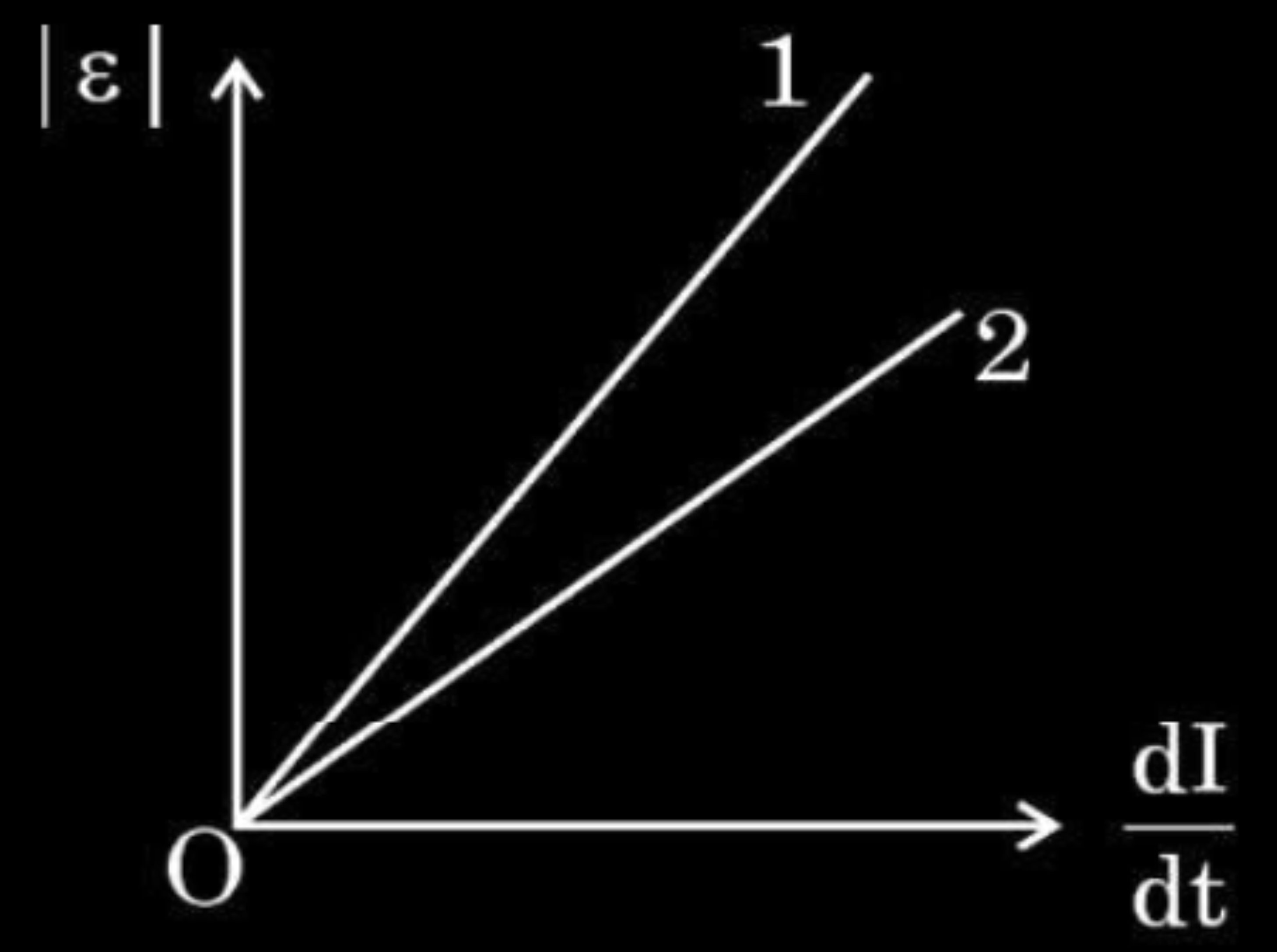
$$\phi = LI$$ ½

$$= 0.1\times\frac{2}{40}\times10$$

$$= 0.05\ Wb$$ ½

24. (a) Obtain an expression for the self-inductance of a long solenoid of length 'l', cross-sectional area 'A' and having 'N' turns.

(b) The figure shows the plot of magnitude of induced emf (ε) versus the rate of change of current in two coils '1' and '2'. Which coil has greater value of self-inductance and why ? *3*



Magnetic field due to a current I flowing in the solenoid $B=\mu_0 nI$ Total flux linked with solenoid is $N\phi_B=(nl)(\mu_0 nI)A$ $=\mu_0 n^2 AlI$ $L=\dfrac{N\phi_B}{I}$ $L=\mu_0 n^2 Al$ (b) Coil 1 Slope of line 1, $\dfrac{\lvert\varepsilon\rvert}{dI/dt}=L$ is more for coil 1 than coil 2	½ ½ ½ ½ **1**

6. When current in a coil changes at a steady rate from 8 A to 6 A in 4 ms, an emf of 1·5 V is induced in it. The value of self-inductance of the coil is :

(A) 6 mH

(B) 12 mH

(C) 3 mH

(D) 9 mH

6. When current in a coil changes at a steady rate from 8 A to 6 A in 4 ms, an emf of 1·5 V is induced in it. The value of self-inductance of the coil is :

(A) 6 mH (B) 12 mH

(C) 3 mH (D) 9 mH

(C) 3mH

5. Two long solenoids of radii r_1 and r_2 ($>r_1$) and number of turns per unit length n_1 and n_2 respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is – **1**

(A) $\dfrac{n_1}{n_2}$

(B) $\dfrac{n_2}{n_1}$

(C) $\dfrac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\dfrac{n_2 r_1^2}{n_1 r_2^2}$

5. Two long solenoids of radii r_1 and r_2 ($>r_1$) and number of turns per unit length n_1 and n_2 respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is – **1**

(A) $\frac{n_1}{n_2}$

(B) $\frac{n_2}{n_1}$

(C) $\frac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\frac{n_2 r_1^2}{n_1 r_2^2}$

(A) $\frac{n_1}{n_2}$

7. A coil of an ac generator, having 100 turns and area 0.1 m^2 each, rotates at half a rotation per second in a magnetic field of 0.02 T. The maximum emf generated in the coil is **1**

(A) 0.31 V (B) 0.20 V

(C) 0.63 V (D) 0.10 V

7. A coil of an ac generator, having 100 turns and area 0.1 m^2 each, rotates at half a rotation per second in a magnetic field of 0.02 T. The maximum emf generated in the coil is 1

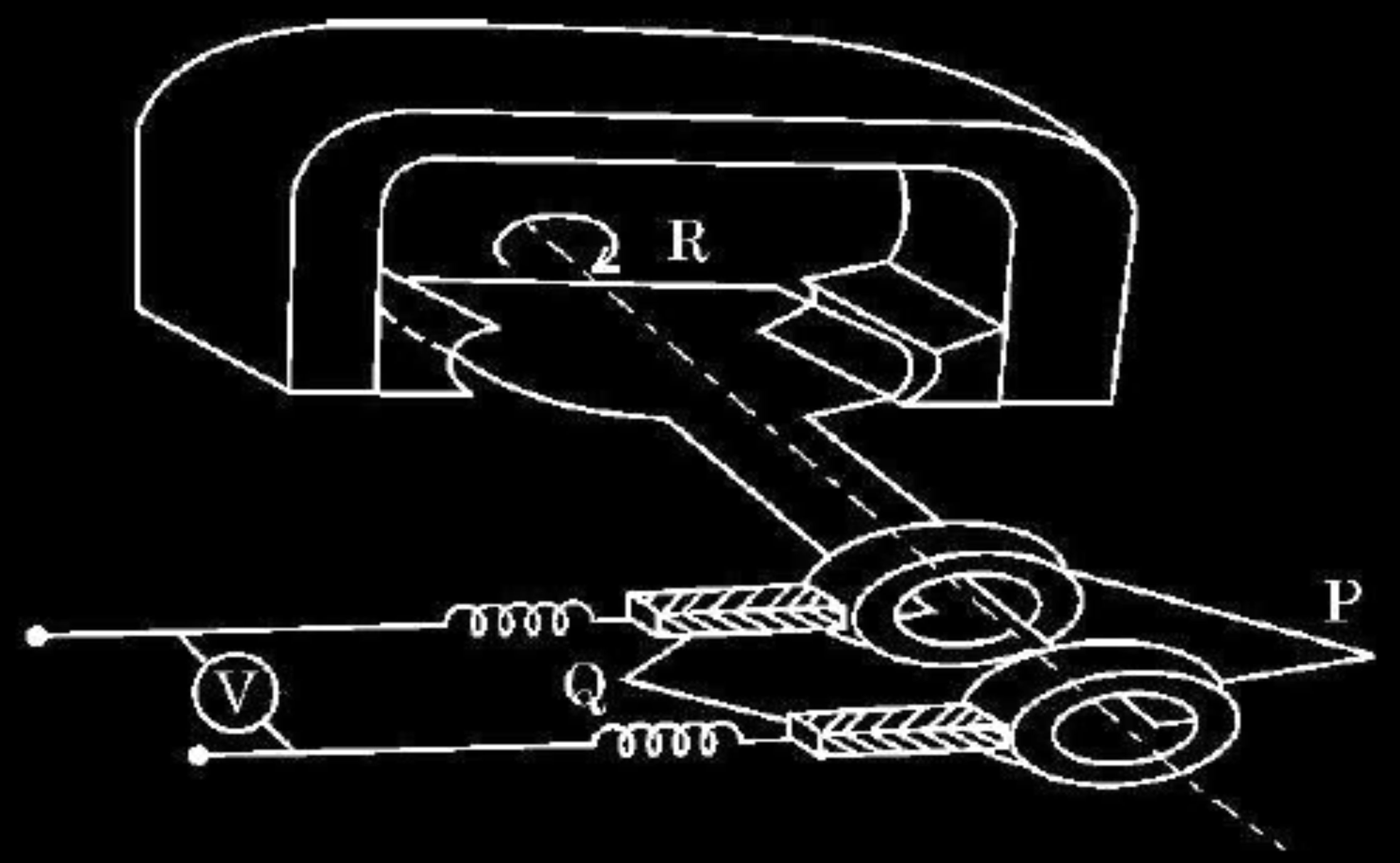
(A) 0.31 V (B) 0.20 V

(C) 0.63 V (D) 0.10 V

(C) 0.63 V

24. (a) State Lenz's law. **3**

(b) In the given figure :



(i) Identify the machine.

(ii) Name the parts P and Q and R of the machine.

(iii) Give the polarities of the magnetic poles.

(iv) Write the two ways of increasing the output voltage.

(a) Lenz's law- The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.	½
(b) (i) AC generator	½
(ii) P – Slip rings Q – Carbon brushes R- Armature coil	½
(iii) Left side of the magnet is North & right side is South or vice-versa.	½
(iv) (Any two) -By increasing the number of turns in the armature coil. -By increasing the speed of rotation of the armature coil. -By increasing the strength of the magnetic field B.	½ + ½

5. You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m. The number of turns in the solenoid should be **1**

(A) 2592

(B) 2866

(C) 2976

(D) 3140

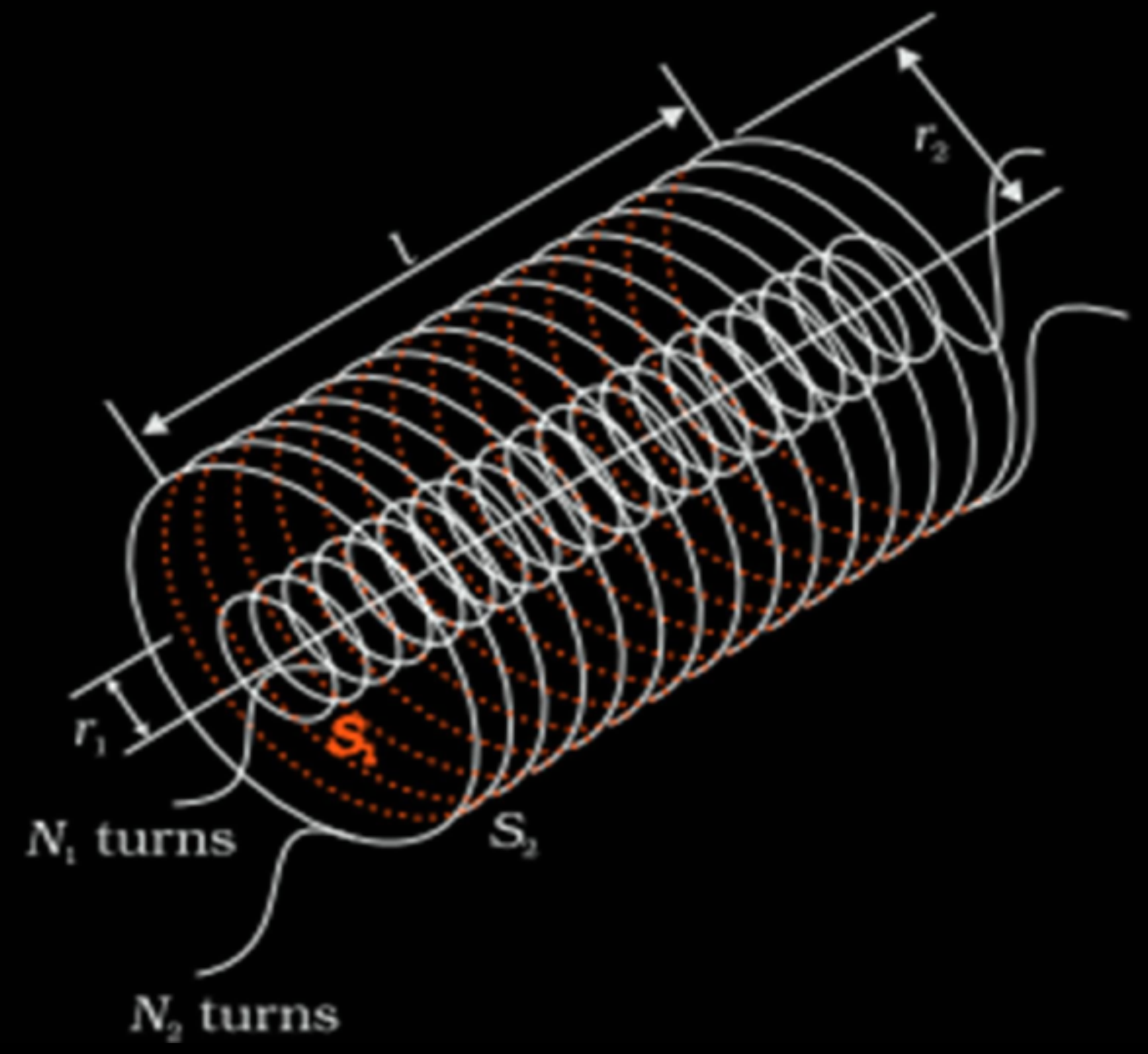
5. You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m. The number of turns in the solenoid should be **1**

(A) 2592 (B) 2866

(C) 2976 (D) 3140

(B) 2866

24. Consider two long co-axial solenoids S_1 and S_2, each of length l ($>> r_2$) and of radius r_1 and r_2 ($r_2 > r_1$). The number of turns per unit length are n_1 and n_2 respectively. Derive an expression for mutual inductance M_{12} of solenoid S_1 with respect to solenoid S_2. Show that $M_{21} = M_{12}$. **3**



½

Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 respectively.
When current I_2 is set up in S_2, flux linkage with solenoid S_1 is –

$$N_1\phi_1 = M_{12}I_2 \quad \text{--(i)}$$ ½

$$N_1\phi_1 = (n_1 l)(\pi r_1^2)(\mu_0 n_2 I_2)$$

$$N_1\phi_1 = \mu_0 n_1 n_2 \pi r_1^2 l\, I_2 \quad \text{--(ii)}$$ ½

From (i) and (ii)

$$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$$ ½

Considering Reverse case, when I_1 current is set up in S_1 , flux linkage with S_2 is –

$$N_2\phi_2 = M_{21}I_1 \quad \text{--(iii)}$$

$$N_2\phi_2 = (n_2 l)(\pi r_1^2)(\mu_0 n_1 I_1) \quad \text{--(iv)}$$ ½

From (iii) and (iv)

$$M_{21} = n_1 n_2 \pi r_1^2 l$$

$$\therefore M_{12} = M_{21}$$ ½

24. (a) Show that the energy required to build up the current I in a coil of inductance L is $\frac{1}{2}LI^2$. **3**

(b) Considering the case of magnetic field produced by air-filled current carrying solenoid, show that the magnetic energy density of a magnetic field B is $\frac{B^2}{2\mu_0}$.

a) Induced emf in an inductor $\lvert\varepsilon\rvert = L\dfrac{dI}{dt}$	½
Rate of work done at any instant $\dfrac{dW}{dt} = \lvert\varepsilon\rvert I$	½
Total Amount of work done in establishing current I $W = \int dW = \int_0^I LIdI$	
Energy required to build up current I is $W = \frac{1}{2} L I^2$	½
b) The Magnetic Energy is $W = U_B = \frac{1}{2} LI^2$ $= \frac{1}{2} L\left(\dfrac{B}{n\mu_0}\right)^2$ as $B = n\,\mu_0 I$	½
Using $L = \mu_0 n^2 Al$ $U_B = \dfrac{1}{2}\left(\mu_0 n^2 Al\right)\left(\dfrac{B^2}{\mu_0^2 n^2}\right)$	½
Energy density $= \dfrac{U_B}{\text{volume}}$ $\dfrac{U_B}{\text{volume}} = \dfrac{1}{2}\times\mu_0 n^2 Al\times\dfrac{B^2}{\mu_0^2 n^2}\times\dfrac{1}{Al}$ $= \dfrac{1}{2}\dfrac{B^2}{\mu_0}$	½

(ii) The currents through two inductors of self-inductance 12 mH and 24 mH are increasing with time at the same rate. Draw graphs showing the variation of the :

(I) magnitude of emf induced with the rate of change of current in each inductor.

(II) energy stored in each inductor with the current flowing through it.

(ii) The currents through two inductors of self-inductance 12 mH and 24 mH are increasing with time at the same rate. Draw graphs showing the variation of the :

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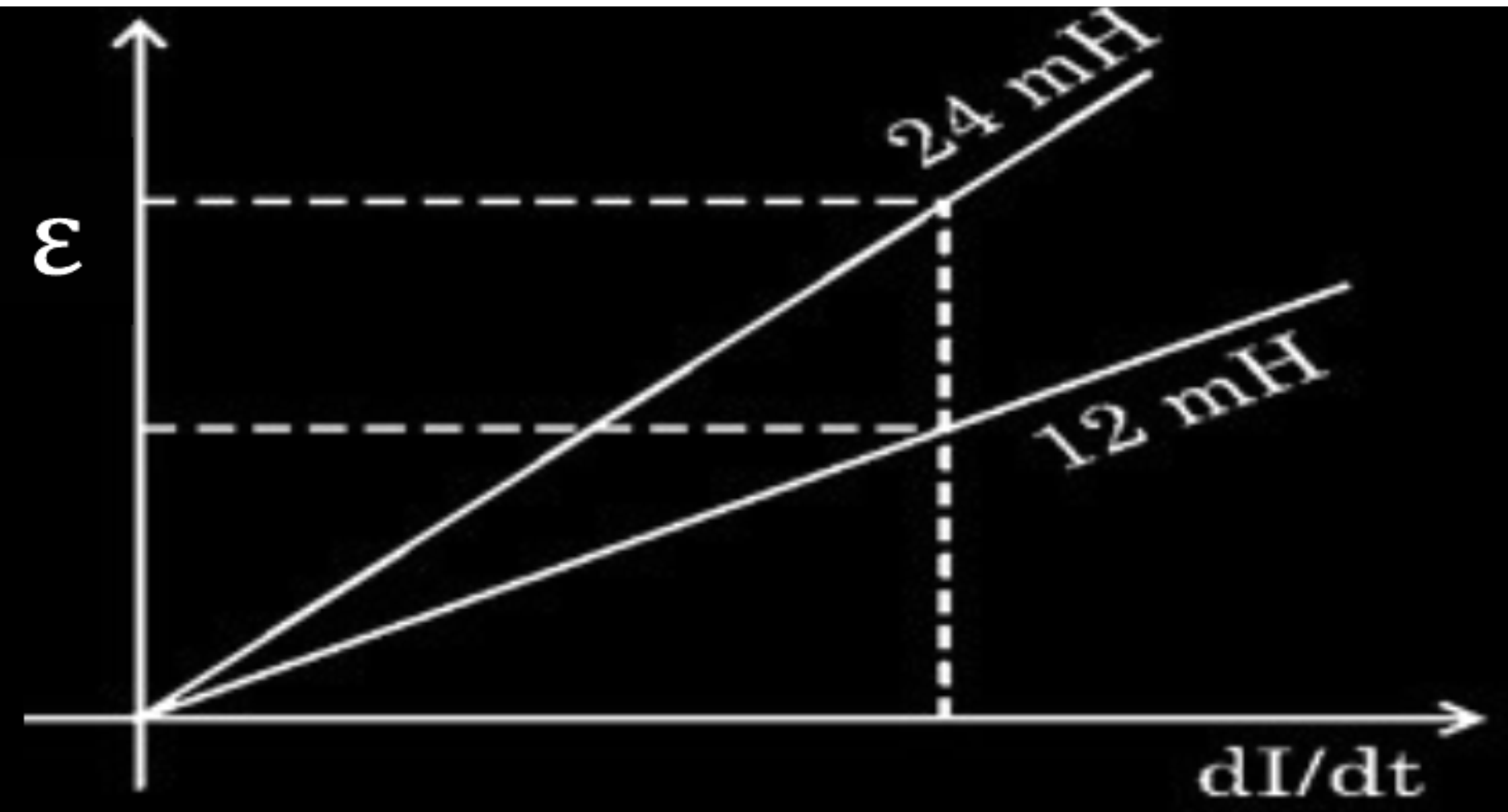

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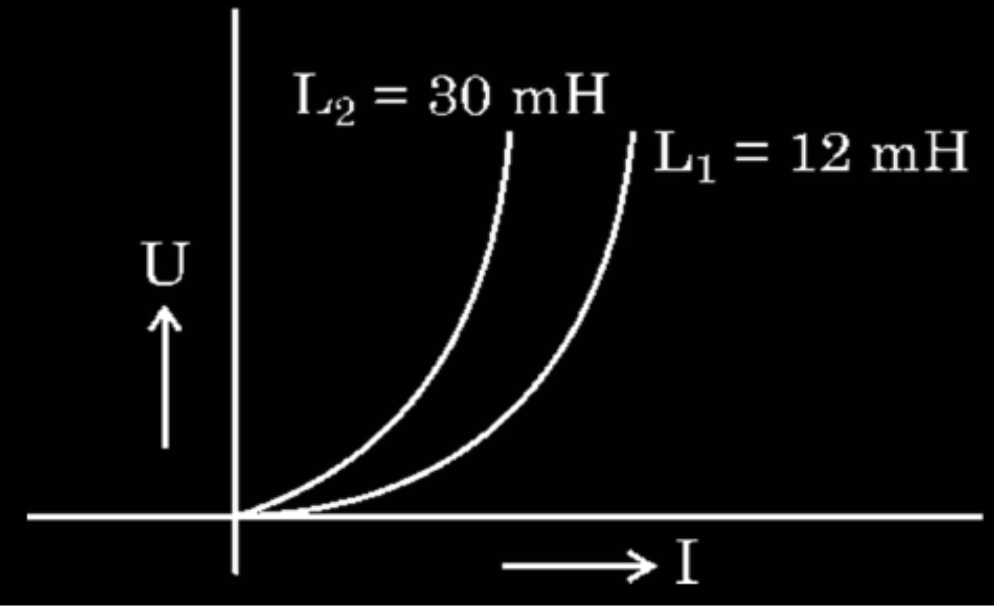
(II) Graph for energy stored



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4. A current of 5·0 A in one coil is reduced to zero in 10 ms inducing an emf of 9·0 V in the neighbouring coil. The mutual inductance of the two coils is :

(A) 18 mH

(B) 1·8 mH

(C) 0·18 mH

(D) 0·18 H

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(A) 18 mH

(B) 1·8 mH

(C) 0·18 mH

(D) 0·18 H

(A)18 mH

6. Which of the following will ***not*** increase the output voltage in a generator ?

(A) Rotating the coil rapidly

(B) Changing the material of the coil

(C) Increasing the area of coil

(D) Increasing the magnetic field through the coil

6. Which of the following will ***not*** increase the output voltage in a generator ?

(A) Rotating the coil rapidly

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(C) Increasing the area of coil

(D) Increasing the magnetic field through the coil

(B) Changing the material of the coil.

23. (a) State Faraday's Law of electromagnetic induction.

(b) A rectangular coil of area A having number of turns N is rotated with angular velocity ω in a uniform magnetic field $\vec{B}$ directly perpendicular to the plane of the coil. Obtain an expression for the instantaneous value of emf induced in the coil. When will the induced emf be maximum ? *3*

23. (a) State Faraday's Law of electromagnetic induction.

(b) A rectangular coil of area A having number of turns N is rotated with angular velocity ω in a uniform magnetic field $\vec{B}$ directly perpendicular to the plane of the coil. Obtain an expression for the instantaneous value of emf induced in the coil. When will the induced emf be maximum ? *3*

(a) The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.	1
(award ½ mark if the student only writes $\varepsilon = -\frac{d\phi_B}{dt}$)	
(b) $\phi = BA \cos\theta = BA \cos\omega t$ For N turns $\phi = NBA \cos\omega t$	½
$\varepsilon = -\frac{d\phi_B}{dt}$	½
$\varepsilon = +NBA\,\omega \sin\omega t$	½
For maximum ε, $\omega t = \frac{\pi}{2}$ $\therefore \varepsilon_0 = NBA\omega$	½

5. The mutual inductance of two coils, in a given orientation is 50 mH. If the current in one of the coils changes as $i = 1.0 \sin\left(100\,\pi t + \frac{\pi}{3}\right)$ A, the peak value of emf (in volt) induced in the other coil will be

(A) $\frac{\pi}{5}$

(B) 5π

(C) 0.5π

(D) 0.05π

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(A) $\frac{\pi}{5}$

(B) 5π

(C) 0.5π

(D) 0.05π

(B) 5π

5. Two coils are placed near each other. When the current in one coil is changed at the rate of 5 A/s, an emf of 2 mV is induced in the other. The mutual inductance of the two coils is

(A) 0.4 mH

(B) 2.5 mH

(C) 10 mH

(D) 2.5 H

5. Two coils are placed near each other. When the current in one coil is changed at the rate of 5 A/s, an emf of 2 mV is induced in the other. The mutual inductance of the two coils is

(A) 0.4 mH (B) 2.5 mH

(C) 10 mH (D) 2.5 H

(A) 0.4 mH

6. A coil of N turns is placed in a magnetic field $\vec{B}$ such that $\vec{B}$ is perpendicular to the plane of the coil. $\vec{B}$ changes with time as $B = B_0 \cos\left(\frac{2\pi}{T}t\right)$ where T is time period. The magnitude of emf induced in the coil will be maximum at **1**

(A) $t = \frac{nT}{8}$

(B) $t = \frac{nT}{4}$

(C) $t = \frac{nT}{2}$

(D) $t = nT$

Here, n = 1, 2, 3, 4, ...

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(A) $t = \frac{nT}{8}$ (B) $t = \frac{nT}{4}$

(C) $t = \frac{nT}{2}$ (D) $t = nT$

Here, n = 1, 2, 3, 4, ...

No option is correct, award 1 mark.

7. A circular loop A of radius R carries a current I. Another circular loop B of radius $r\left(=\frac{R}{20}\right)$ is placed concentrically in the plane of A. The magnetic flux linked with loop B is proportional to **1**

(A) R (B) $\sqrt{R}$

(C) $R^{\frac{3}{2}}$ (D) R^2

7. A circular loop A of radius R carries a current I. Another circular loop B of radius $r\left(=\frac{R}{20}\right)$ is placed concentrically in the plane of A. The magnetic flux linked with loop B is proportional to **1**

(A) R (B) $\sqrt{R}$

(C) $R^{\frac{3}{2}}$ (D) R^2

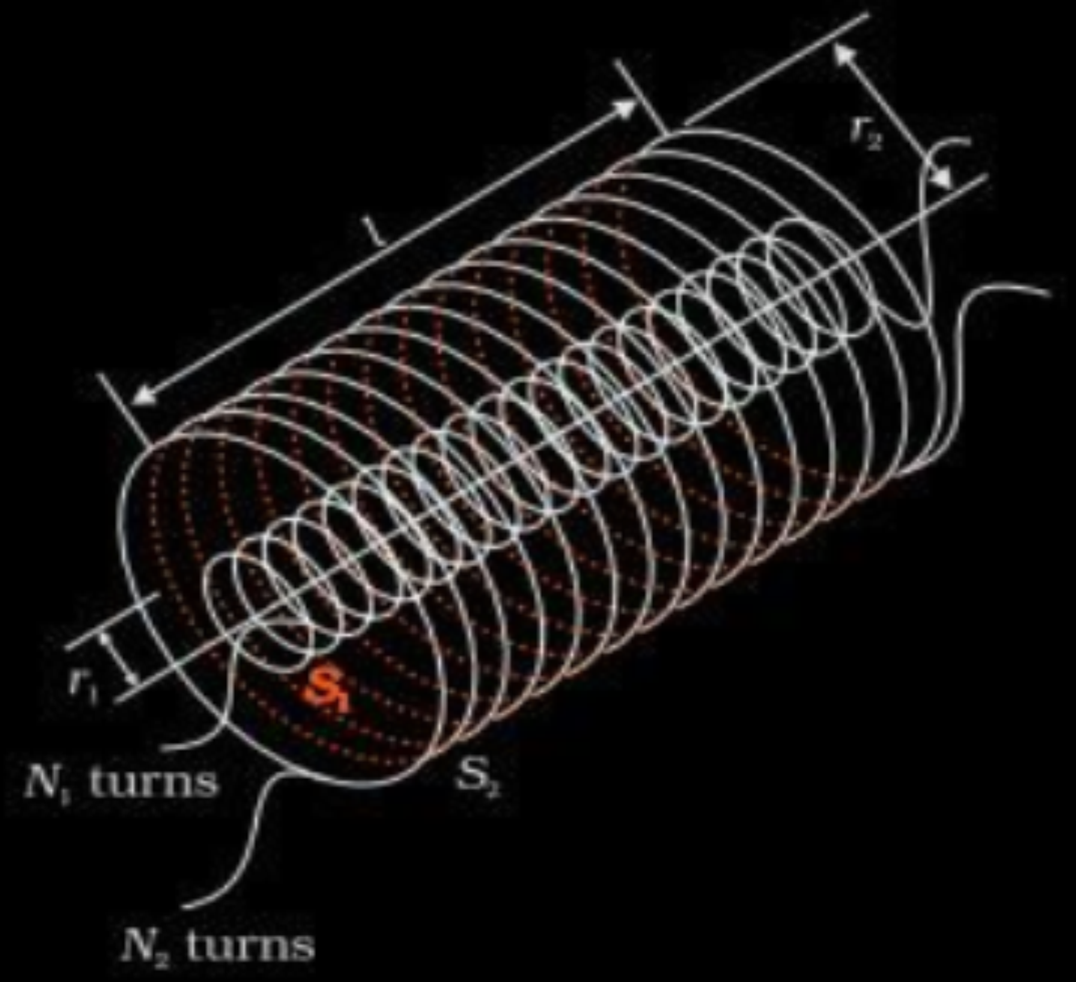
(A) R

22. (a) (i) Define mutual inductance. Write its SI unit. **3**

(ii) Derive an expression for the mutual inductance of a system of two long coaxial solenoids of same length l, having turns N_1 and N_2 and of radii r_1 and r_2 ($> r_1$).

(i) Mutual inductance between two coils is defined as the magnetic flux associated with a coil when unit current flows through neighbouring coil.	½
Alternatively Mutual inductance between two coils is defined as the magnitude of induced emf in a coil when the rate of change of current in neighbouring coil is unity.	
SI unit of mutual inductance is henry(H).	½

(ii)



When current I_2 flows in outer solenoid, the resulting flux linkage with inner solenoid.

$$N_1\phi_1 = N_1B_2A_1$$ ½

$$N_1\phi_1 = N_1\left(\frac{\mu_0 N_2 I_2}{l}\right)\pi r_1^2$$

$$N_1\phi_1 = \frac{\mu_0 N_1 N_2 \pi r_1^2 I_2}{l} \quad \text{..........................(1)}$$ ½

$$N_1\phi_1 = M_{12}I_2 \quad \text{.......................... (2)}$$ ½

From equations (1) and (2)

$$M_{12} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$ ½

7. Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if :

(A) both l and A are increased

(B) l is decreased and A is increased

(C) l is increased and A is decreased

(D) both l and A are decreased

7. Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if :

(A) both l and A are increased

(B) l is decreased and A is increased

(C) l is increased and A is decreased

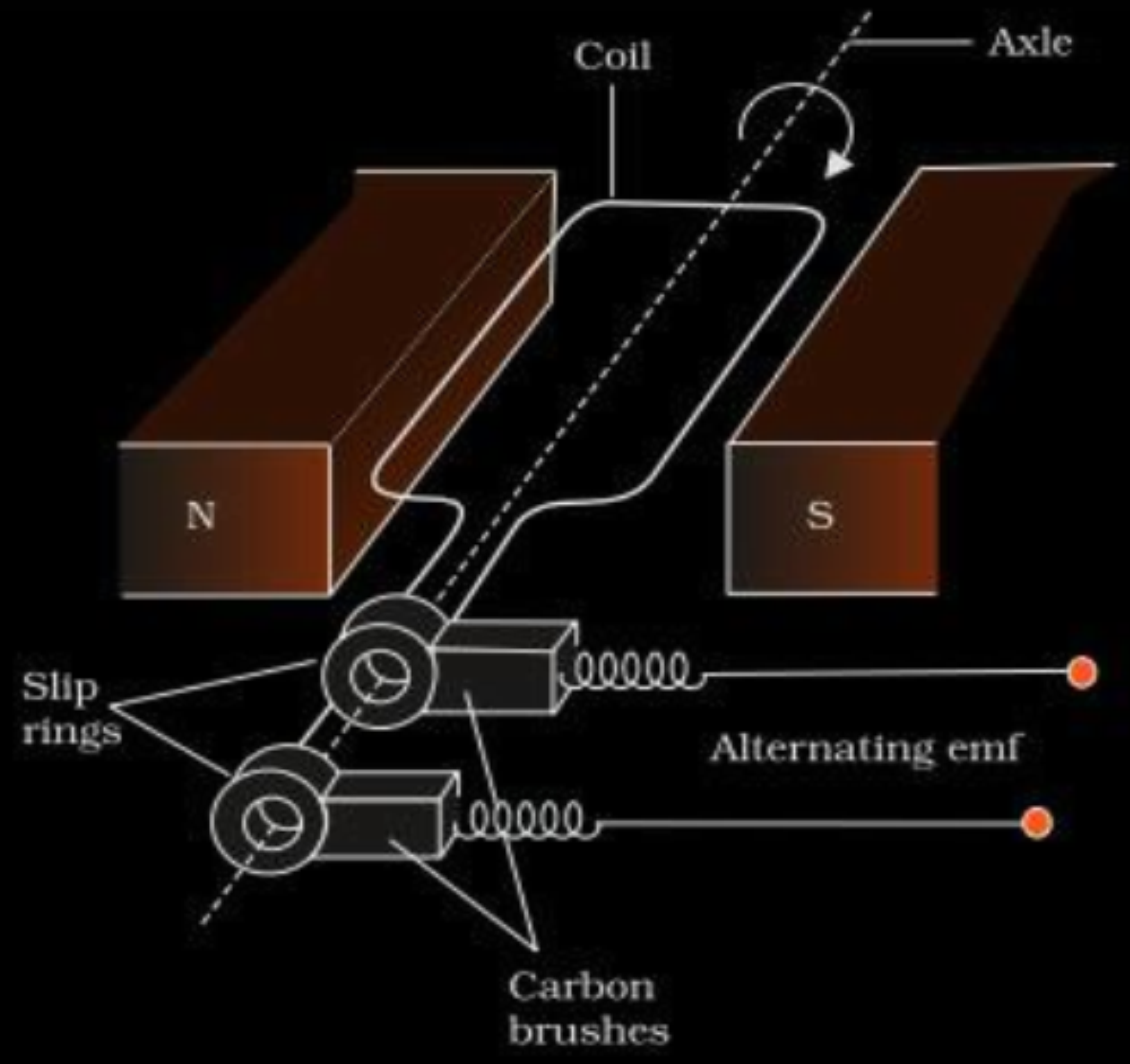
(D) both l and A are decreased

(B) l is decreased and A is increased

(b) (i) A rectangular coil of N turns and area of cross-section A is rotated at a steady angular speed ω in a uniform magnetic field. Obtain an expression for the emf induced in the coil at any instant of time.

(ii) Two coplanar and concentric circular loops L_1 and L_2 are placed coaxially with their centres coinciding. The radii of L_1 and L_2 are 1 cm and 100 cm respectively. Calculate the mutual inductance of the loops. (Take $\pi^2 = 10$) *5*

(i) The flux at any instant t is	
$\phi = NBA\cos\theta = NBA\cos\omega t$	½
From Faraday's law $\boldsymbol{\varepsilon = -\frac{d\phi_B}{dt}}$	½
$= -NBA\frac{d}{dt}(\cos\omega t)$	½
$\boldsymbol{\varepsilon} = -NBA\,\omega\sin\omega t$	½
(ii) $M = \frac{\mu_0 \pi r_1^2}{2r_2} = \frac{4\pi\times10^{-7}\times\pi r_1^2}{2r_2}$	½+½
$= \frac{2\times10\times10^{-7}\times(10^{-2})^2}{100\times10^{-7}}$	½
$= 2\times10^{-10}$H	½



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6. The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1, the current changes from 4 A to zero in $0 \cdot 2$ s. If the resistance of coil C_2 is 4 Ω, then the charge that flows through it per second will be :

(A) $4 \cdot 0$ C (B) $1 \cdot 5$ C

(C) $0 \cdot 05$ C (D) $0 \cdot 1$ C

6. The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1, the current changes from 4 A to zero in 0·2 s. If the resistance of coil C_2 is 4 Ω, then the charge that flows through it per second will be :

(A) 4·0 C

(B) 1·5 C

(C) 0·05 C

(D) 0·1 C

(D) 0.1 C

6. The current in a coil of 15 mH increases uniformly from zero to 4 A in 0·004 s. The emf induced in the coil will be :

(A) 22·5 V
(B) 17·5 V
(C) 15·0 V
(D) 12·5 V

6. The current in a coil of 15 mH increases uniformly from zero to 4 A in 0·004 s. The emf induced in the coil will be :

(A) 22·5 V (B) 17·5 V

(C) 15·0 V (D) 12·5 V

(C) 15.0 V

3. The emf induced in a coil rotating in a magnetic field does ***not*** depend upon the following :

(A) Area of the coil

(B) Resistance of the coil

(C) Number of turns in the coil

(D) Angular speed of rotation of the coil

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(A) Area of the coil

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(C) Number of turns in the coil

(D) Angular speed of rotation of the coil

(B) Resistance of the coil

15. *Assertion (A) :* The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R) : The flux linkage between two coils is maximum when they are wound on each other.

15. *Assertion (A) :* The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R) : The flux linkage between two coils is maximum when they are wound on each other.

(A) Assertion(A) is true , Reason (R) is true and Reason (R) is correct explanation of the Assertion (A)

32. (a) (i) Define the term 'self-inductance' of a coil. Obtain an expression for self-inductance of a solenoid of area of cross-section A, length L and having n turns per unit length.

(ii) The current flowing in a coil of inductance 50 mH is reduced from 10 A to 0 in 20 ms. Calculate the change in flux linked with the coil. *5*

(i) Self-Inductance is the flux linked with a coil when unit current flows through it. Alternatively: Self -Inductance is the emf induced in a coil when the rate of change of current through the coil is unity.	1
Magnetic field due to current I flowing through the solenoid $B=\mu_0 nI$	½
Flux linked with the solenoid $\phi=\vec{B}.\vec{A}$ $=\mu_0 nIA$	½
Total flux linked with the solenoid $=(nL)(\mu_0 nI)A$	½
$=\mu_0 n^2 ALI$	½
$\therefore$ Self –Inductance $=\mu_0 n^2 AL$	
(ii) $\Delta\phi = L\Delta I$	1
$=50\times10^{-3}(10)$	½
$=0.5$ Wb	½

(b) (i) Briefly explain the working of an ac generator. Obtain the expression for the instantaneous value of the emf induced in the generator.

(i)The armature coil is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change so an emf is induced in it. Magnetic flux linked with the coil in orientation θ. $\phi = NBA\cos\theta$ $= NBA\cos\omega t$ $e = -\dfrac{d\phi}{dt}$ $= NBA\omega\sin\omega t$ $e = e_0\sin\omega t$	1 ½ ½ ½ ½

(i) With the help of a labelled diagram, explain the working of an ac generator. Obtain the expression for the emf induced at an instant ‘t’.

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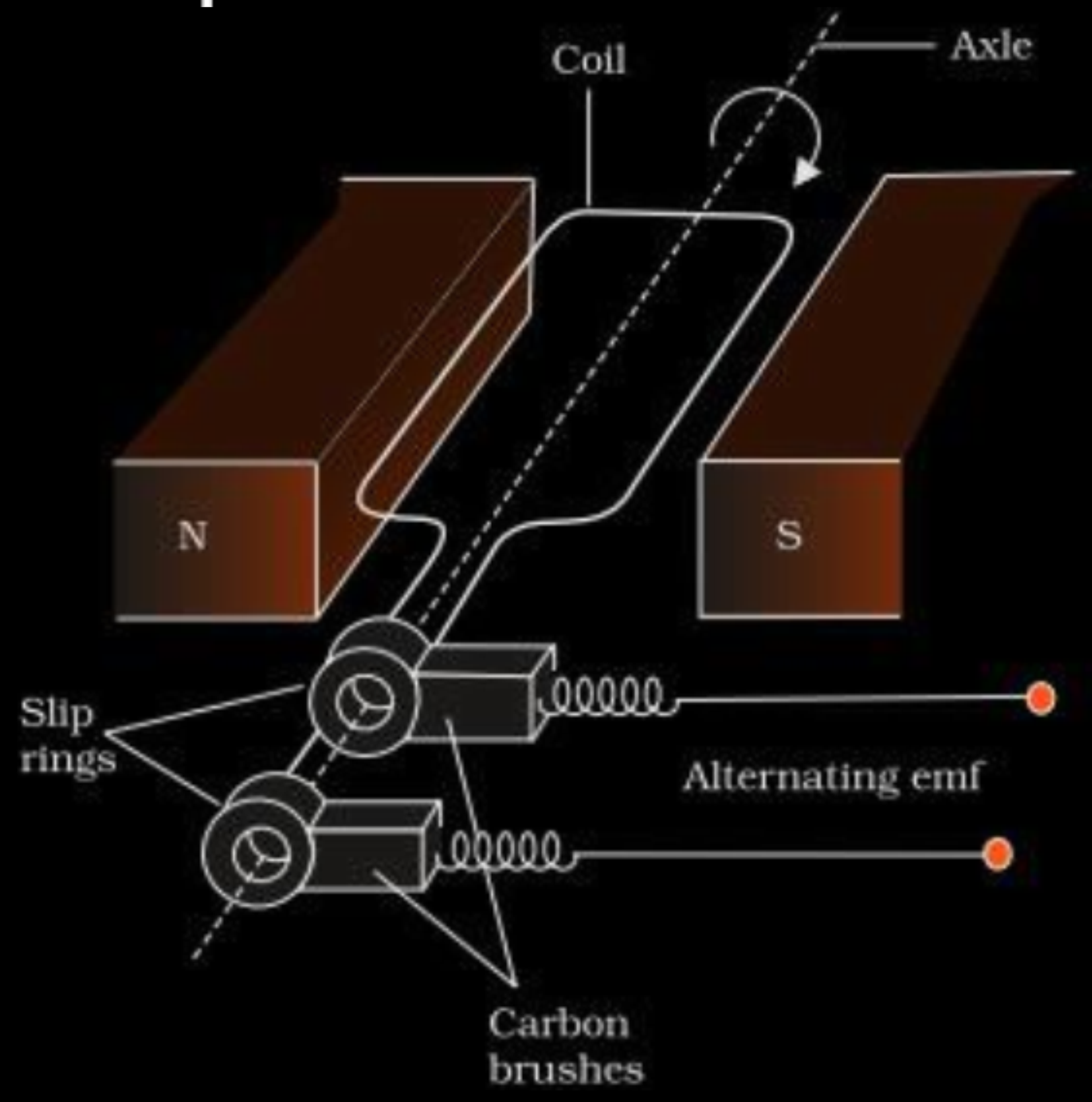

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Working of ac generator

When coil is rotated in a uniform magnetic field with a constant angular speed ω, magnetic flux through it changes. As a result, an e.m.f is induced in the coil. 1

Flux linked with the coil at any instant 't' is

$$\varphi_B = BA\cos\omega t$$ ½

$$\varepsilon = -N\frac{d\varphi_B}{dt}$$

$$\varepsilon = NBA\omega\sin\omega t$$ ½

(i) State Faraday's law of electromagnetic induction and mention the utility of Lenz's law. Obtain an expression for self-inductance of a coil in terms of its geometry and permeability of the medium.

(i) The magnitude of induced e.m.f in a circuit is equal to the time rate of change of magnetic flux through the circuit	½
Utility of Lenz's law	
It give polarity of the induced e.m.f .	½
Expression for self inductance	
Consider a long solenoid of cross-sectional area A and length l, having n turns per unit length. If I is the current flowing in the solenoid, magnetic field inside the solenoids is	
$B = \mu_0 nI$	½
Total magnetic flux linked with the solenoid is	
$N\varphi_B = (nl)(\mu_0 nI)(A)$	
$N\varphi_B = \mu_0 n^2 AlI$	½
Self inductance	
$L = \frac{N\varphi_B}{I}$	
$L = \mu_0 n^2 Al$	½
If solenoid is filled with a material of relative permeability μ_r , then	
$L = \mu_r \mu_0 n^2 Al$	½

23. (a) Define coefficient of self-inductance.

(b) A solenoid has length 12 cm, area of cross-section 24 cm^2 and 1500 turns. Calculate the self-inductance of the solenoid. *3*

23. (a) Define coefficient of self-inductance.

(b) A solenoid has length 12 cm, area of cross-section 24 cm^2 and 1500 turns. Calculate the self-inductance of the solenoid. *3*

(a) Coefficient of self – induction – It is the flux linked with a coil when the current flowing through the coil is unity.	1
Alternatively- Coefficient of self – induction – It is the self induced emf in a coil when the current in the coil changes at the rate of 1 A/S	
(b) Coefficient of self induction (L)	
$L=\dfrac{N^2\mu_0 A}{l}$	½
$L=\dfrac{(1500)^2\times 4\pi\times 10^{-7}\times 24\times 10^{-4}}{12\times 10^{-2}}$	½
$L=18\pi\times 10^{-3}H$	
$=5.65\times 10^{-2}$ H	1

24. A coil of area of cross-section 'A' has N number of turns. It is rotating with an angular speed ω inside a uniform magnetic field B. If the total resistance of circuit of coil is R, find : *3*

(a) the maximum current in the circuit.

(b) the flux through the coil when the current is zero.

(c) the flux through the coil when the current is maximum.

(a) Induced $Emf\,(\varepsilon) = N\dfrac{-d\phi_B}{dt} = -NBA\dfrac{d}{dt}(\cos\omega t)$ Induced $emf\,(\varepsilon) = NBA\,\omega\,sin\omega\,t$ $Current(\mathrm{I}) = \dfrac{\varepsilon}{R} = \dfrac{NBA\omega Sin\omega t}{R}$ Max current $(I_{\max}) = \dfrac{NBA\omega}{R}$	1
(b) Flux through the coil $\phi_B = NBACos\omega t$ When current is zero, $\omega t = 0^0$ $\phi_B = NBA$	 1
(c) When current is maximum, $\omega t = 90^0$ $\phi_B = 0$	1

8. Which of the following statements is ***not*** correct ?

(a) Lenz's law is the consequence of the law of conservation of energy.

(b) The magnitude of induced emf in a coil is directly proportional to the rate of change of magnetic flux.

(c) Inserting an iron core in a coil decreases its self-inductance.

(d) An emf can be induced in a straight conductor by moving it perpendicularly through a uniform magnetic field.

8. Which of the following statements is ***not*** correct ?

(a) Lenz's law is the consequence of the law of conservation of energy.

(b) The magnitude of induced emf in a coil is directly proportional to the rate of change of magnetic flux.

(c) Inserting an iron core in a coil decreases its self-inductance.

(d) An emf can be induced in a straight conductor by moving it perpendicularly through a uniform magnetic field.

(c) Inserting an iron core in a coil decrease its self inductance

28. Define the term 'self-inductance' of a coil. Obtain the expression for the self-inductance of a long solenoid of length l, cross-sectional area A and having N turns. *3*

28. Define the term 'self-inductance' of a coil. Obtain the expression for the self-inductance of a long solenoid of length l, cross-sectional area A and having N turns. *3*

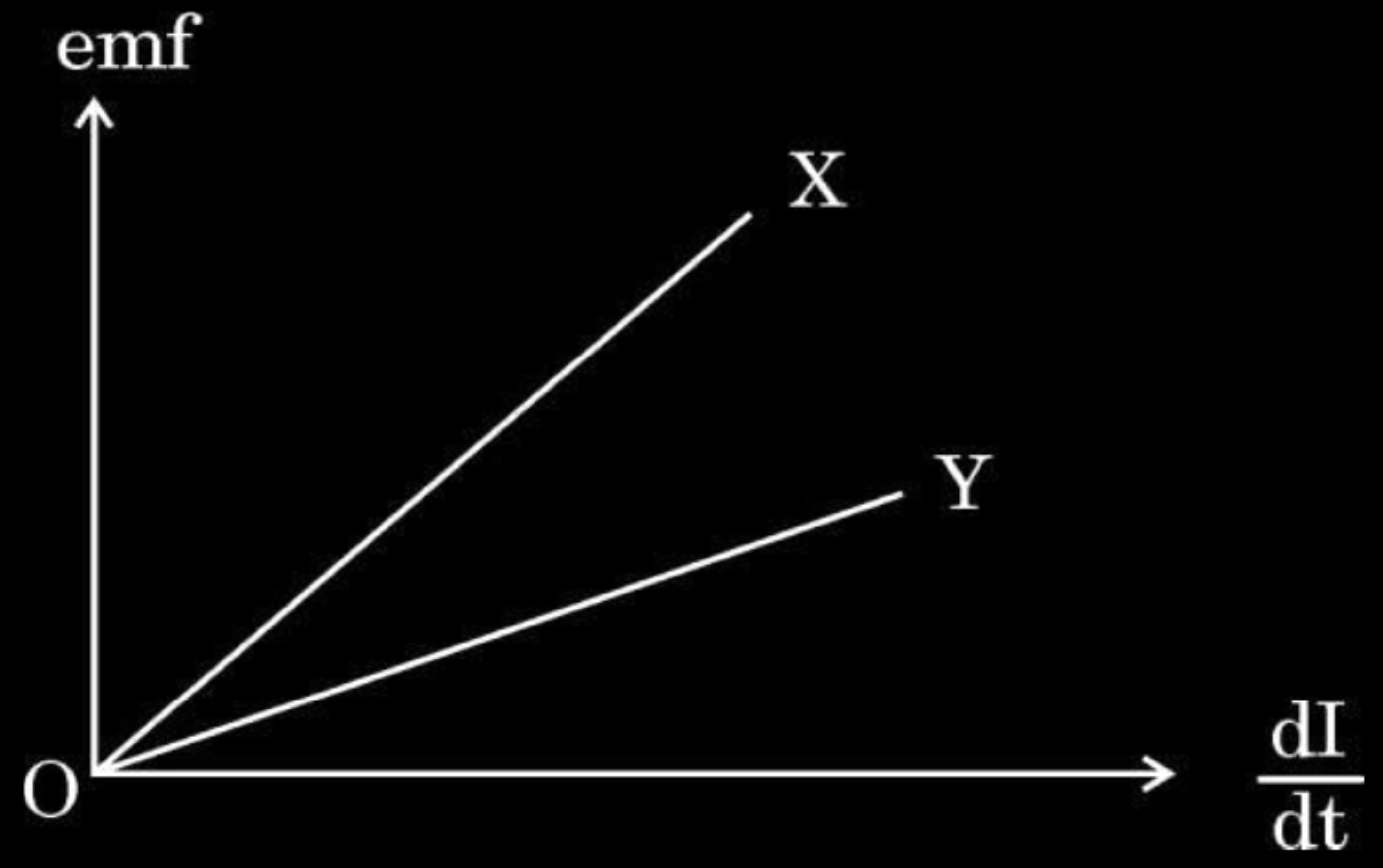
It is defined as magnetic flux through a solenoid when unit current passes through the solenoid. $$L = \frac{\phi}{I}$$	1
Flux linkage through coil of N turns $$N\phi_B \propto I \qquad \_\_\_\_\_\_\text{(i)}$$	½
$$N\phi_B = LI \qquad \_\_\_\_\_\_\text{(ii)}$$	
Induced emf $\varepsilon = -N\dfrac{d\phi_B}{dt} = -NL\dfrac{dI}{dt}$ ______(iii)	½
$$N\phi_B = N\frac{A\mu_o NI}{l} \qquad \_\_\_\_\_\_\text{(iv)}$$	½
From eq. (ii) and (iv) $$L = \mu_o \frac{N^2 A}{l} \qquad \_\_\_\_\_\_\text{(v)}$$	½

30. (a) Two coplanar concentric circular loops of radii R and r (<< R) are arranged coaxially. Obtain the expression for their mutual inductance.

30. (a) Two coplanar concentric circular loops of radii R and r (<< R) are arranged coaxially. Obtain the expression for their mutual inductance.

(a) Magnetic field due to current I flowing in the loop with radius R, at its centre. $$B=\frac{\mu_0 I}{2R}$$	½
Magnetic flux linked with the smaller coil $$\phi=\frac{\mu_0 I}{2R}.\pi r^2$$	½
which is equal to MI	½
hence $M=\dfrac{\mu_0 \pi r^2}{2R}$	½

27. (a) The figure shows the variation of induced emf as a function of rate of change of current for two identical solenoids X and Y. One is air cored and the other is iron cored. Which one of them is iron cored ? Why ?



(b) Obtain an expression for self-inductance of a long solenoid of length L and cross-sectional area A having N turns. *3*

(a) X is iron cored.	½
Reason- From the graph-	
Slope of the graph $= \dfrac{\varepsilon}{\left\| dI/dt \right\|} = L$	
Slope of X is more than that of Y. Hence X is iron cored because Inductance of iron cored coil is more than that of air cored coil.	½
(b) Magnetic field due to solenoid (axis)	
$B = \dfrac{\mu_0 N I}{l}$	½
Magnetic flux through the solenoid	
$\phi_B = N\phi = \dfrac{\mu_0 N^2 A}{L} I$	½
Since self-inductance $= \dfrac{\phi_B}{I}$	½
$= \dfrac{\mu_0 N^2 A}{L}$	½

27. What is meant by the term 'mutual inductance' of a pair of coils ? Obtain an expression for the mutual inductance of two long coaxial solenoids, each of length l but having different number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 > r_1$). *3*

<u>Mutual Inductance</u>-It is the magnetic flux in the secondary coil due to the flow of unit current in the primary coil.	1
Alternatively:	
It is the emf induced in the secondary coil when rate of change of current in the primary coil is unity.	
<u>Expression for Mutual Inductance</u>	
Let N_1 and N_2 be the total number of turns of coils S_1 &S_2, respectively.	
When a current I_2 is set up through S_2, it in turn sets up a magnetic flux through $S_1 \Rightarrow N_1\phi_1 = M_{12}I_2$ ---------(1)	½
M_{12} is the mutual inductance of solenoid S_1 w.r.t. S_2.	
Magnetic field due to I_2 in S_2 $= \mu_o(\frac{N_2}{l})I_2$	½
$N_1\phi_1 = (N_1)(\pi r_1^2)(\mu_o \frac{N_2}{l} I_2)$ ---------(2)	½
From equation (1) & equation (2)	
$M_{12} = \frac{\mu_o N_1 N_2 \pi r_1^2}{l}$	½

9. A circular coil of radius 8·0 cm and 40 turns is rotated about its vertical diameter with an angular speed of $\frac{25}{\pi}$ rad s^{-1} in a uniform horizontal magnetic field of magnitude $3{\cdot}0 \times 10^{-2}$ T. The maximum emf induced in the coil is :

(a) 0·12 V (b) 0·15 V

(c) 0·19 V (d) 0·22 V

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(a) 0·12 V (b) 0·15 V

(c) 0·19 V (d) 0·22 V

(c) 0.19 V

28. A long solenoid of radius r consists of n turns per unit length. A current $I = I_0 \sin \omega t$ flows in the solenoid. A coil of N turns is wound tightly around it near its centre. What is :

(a) the induced emf in the coil ?

(b) the mutual inductance between the solenoid and the coil ? *3*

a) magnetic field produced in the solenoid near the center	
$B = \mu_o n I$	½
Flux linked with the coil wound over solenoid	
$\phi = NBA = N \pi r^2 B$	
$= N \pi r^2 \mu_o n I$	½
Induced emf $e = \frac{-d\phi}{dt} = -\pi r^2 N n \mu_o \frac{dI}{dt}$ (i)	½
$= - \mu_o \pi r^2 n N I_o \omega \cos \omega t$	½
b) comparing Eq (i) with $e = -M \frac{dI}{dt}$	½
$M = \mu_o \pi r^2 n N$	½

31. (a) (i) Define coefficient of self-induction. Obtain an expression for self-inductance of a long solenoid of length l, area of cross-section A having N turns.

(ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $\left(\frac{200}{\pi}\right)$ Hz and a DC source is applied across the coil. **5**

AC Source		
S.No.	V (Volts)	I (A)
1	3.0	0.5
2	6.0	1.0
3	9.0	1.5

DC Source		
S.No.	V (Volts)	I (A)
1	4.0	1.0
2	6.0	1.5
3	8.0	2.0

(i) Coefficient of self induction is defined as the amount of magnetic flux associated with a coil when unit current flows through it.	1
Alternatively	
It is defined as the magnitude of emf induced in a coil when current changes at the rate of 1 A/s through it.	
(ii) The magnetic field due to a current I flowing in solenoid is	
$B = \dfrac{\mu_0 NI}{l}$	½
The total magnetic flux linked with solenoid	
$N\phi_B = (N)\left(\dfrac{\mu_0 NI}{l}\right)(A)$	
$= \dfrac{\mu_0 N^2 IA}{l}$	½
The self inductance is	
$L = \dfrac{N\phi_B}{I}$	½
$L = \dfrac{\mu_0 N^2 A}{l}$	½

(iii)From the table , Z=6 Ω , R = 4Ω $Z^2 = R^2 + X_L^2$ $X_L^2 = Z^2 - R^2 = 36 - 16 = 20$ $X_L = 2\sqrt{5} \approx 4.5\ \Omega$ $2\pi\nu L = 4.5$ $L = \dfrac{4.5}{2\times\pi\times\dfrac{200}{\pi}}$ $L = 1.1\times10^{-2} H = 11 mH$ Note : Please do not deduct marks if a student writes answer as $0.5\sqrt{5}\times10^{-2}$ H	½ ½ ½ ½

(b) (i) With the help of a labelled diagram, describe the principle and working of an ac generator. Hence, obtain an expression for the instantaneous value of the emf generated.

(ii) The coil of an ac generator consists of 100 turns of wire, each of area 0.5 m^2. The resistance of the wire is 100 Ω. The coil is rotating in a magnetic field of 0.8 T perpendicular to its axis of rotation, at a constant angular speed of 60 radian per second. Calculate the maximum emf generated and power dissipated in the coil.

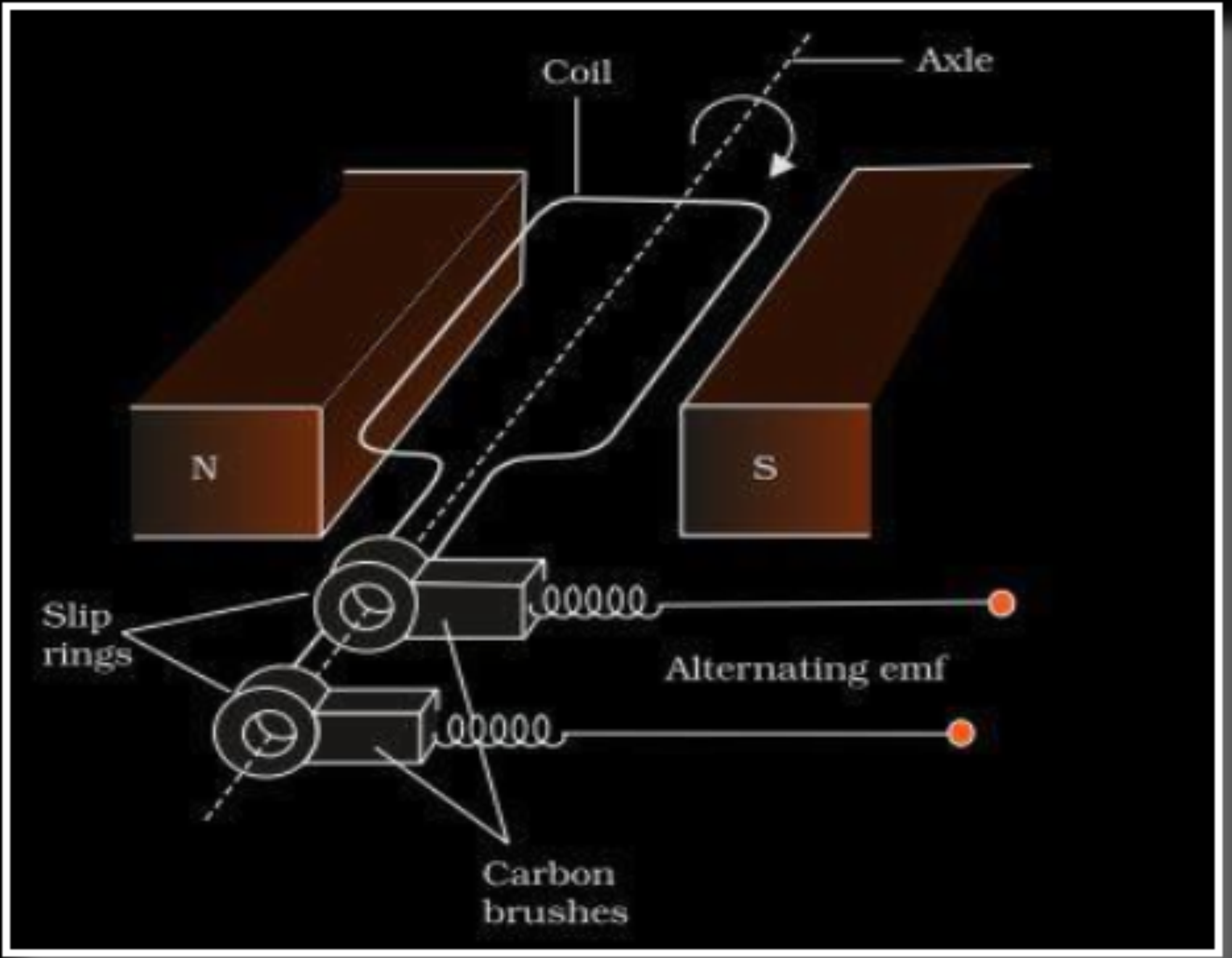
Principle – It is based on the principle of electromagnetic induction.

Whenever there is a change in magnetic flux linked with a coil , an emf is induced in the coil. ½

Alternatively

Give full credit if a candidate writes ,it is based on the principle of electromagnetic induction.

Working - When a rectangular coil is rotated in a magnetic field, the magnetic flux changes continuously which induces an emf and the direction of current changes periodically. ½



1

$$\varepsilon = \frac{-Nd\phi}{dt}$$ ½

$$= -NBA\frac{d}{dt}(\cos\omega t)$$

$$\varepsilon = NBA\omega\sin\omega t$$ ½

(ii) $\varepsilon_0 = NBA\omega$ ½

$$= 100\times0.8\times0.5\times60$$

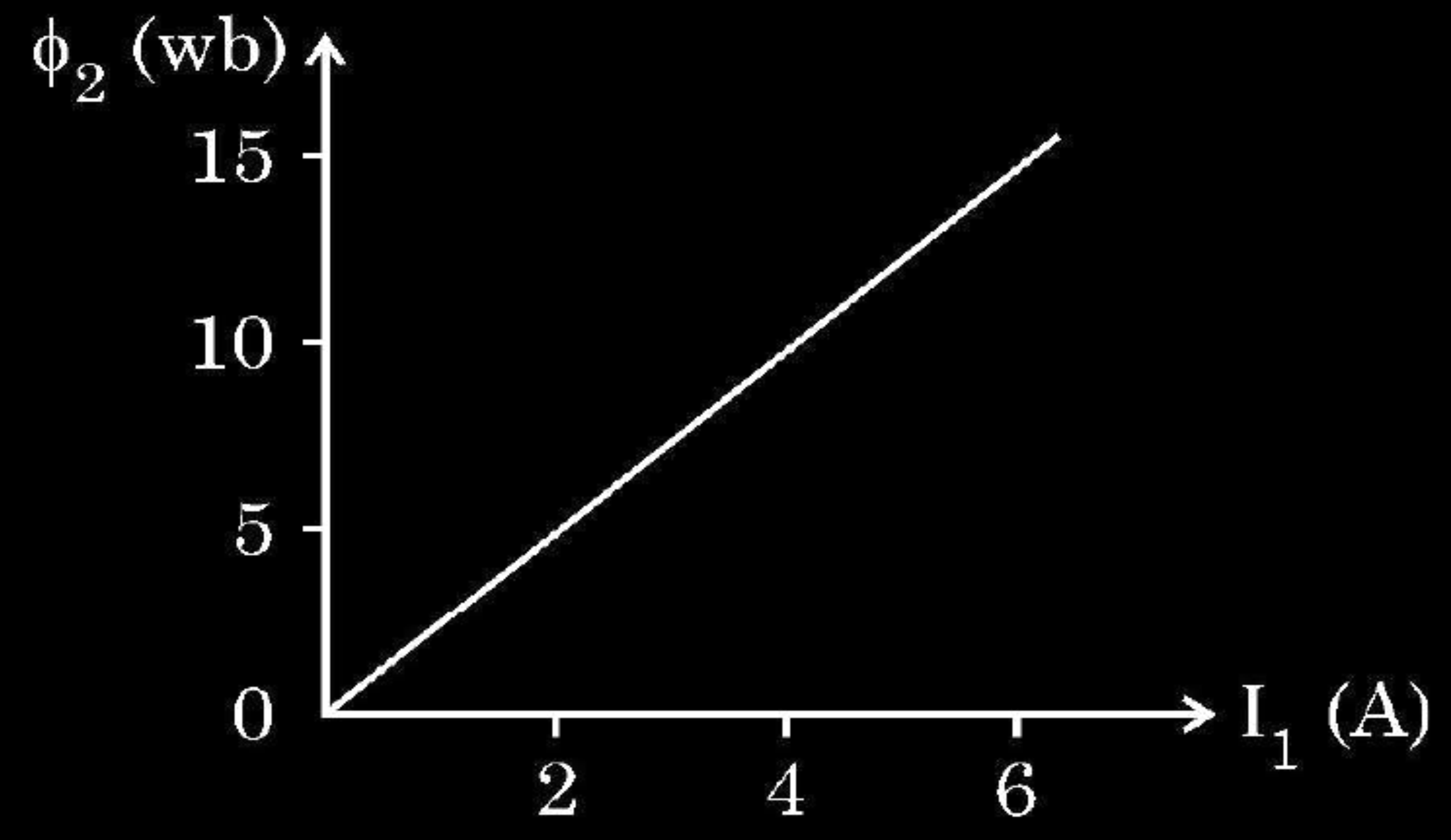
$$= 2400\text{ V}$$ ½

Power dissipated , $P = \dfrac{\varepsilon_{rms}^2}{R}$ ½

$$= \frac{\left(\frac{2400}{\sqrt{2}}\right)^2}{100}$$

$$= 28.8\text{ kW}$$ ½

25. Two coils C_1 and C_2 are placed close to each other. The magnetic flux ϕ_2 linked with the coil C_2 varies with the current I_1 flowing in coil C_1, as shown in the figure. Find **2**



(i) the mutual inductance of the arrangement, and

(ii) the rate of change of current $\left(\frac{dI_1}{dt}\right)$ that will induce an emf of 100 V in coil C_2.

(i)	$\phi_2 = MI_1$ Slope of the graph , M = $\dfrac{\phi_2}{I_1}$ M = 2.5 H	½ ½
(ii)	$\left\|\in_2\right\| = M\dfrac{dI_1}{dt}$ $\dfrac{dI_1}{dt} = \dfrac{\left\|\in_2\right\|}{M} = \dfrac{100}{2.5} = 40\,\mathrm{A}s^{-1}$	½ ½

28. State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced. **3**

28. State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced. **3**

Principle: Whenever magnetic flux linked through a coil changes an emf is induced.	1
Alternatively: ac generator is based on the phenomenon of electromagnetic induction.	
Working: The coil (armature) is mechanically rotated in the uniform magnetic field by some external means. The rotation of the coil causes the magnetic flux through it to change. So an emf is induced in the coil.	1
Expression for induced emf Flux linked with the coil at any instant of time is- $\phi_B = BA\cos\omega t$ $\in = -N\frac{d\phi_B}{dt}$ $\in = -NBA\frac{d}{dt}(\cos\omega t)$	½
Thus, instantaneous value of emf is $\in = NBA\omega\sin\omega t$	½

19. A closely wound coil having 100 turns carries a current of 2 A. If the magnetic flux linked with coil is $\pi^2 \times 10^{-5}$ Wb, find the radius of the coil. *2*

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$\phi = BNA$	½
$B = \dfrac{\mu_O IN}{2r}$	½
$\phi = N \times \dfrac{\mu_O NI}{2r} \times \pi r^2$	
$\Rightarrow \pi^2 \times 10^{-5} = \dfrac{4\pi \times 10^{-7} \times (100)^2 \times 2 \times r^2}{2r}$	½
r = 0.25 cm	½

13. A planar loop is rotated in a magnetic field about an axis perpendicular to the field. The polarity of induced emf changes once in each :

(a) 1 revolution

(b) $\left(\frac{1}{2}\right)$ revolution

(c) $\left(\frac{1}{4}\right)$ revolution

(d) $\left(\frac{3}{4}\right)$ revolution

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(b) $(\frac{1}{2})$ revolution

19. Plot a graph showing variation of induced e.m.f. with the rate of change of current flowing through a coil. **1**

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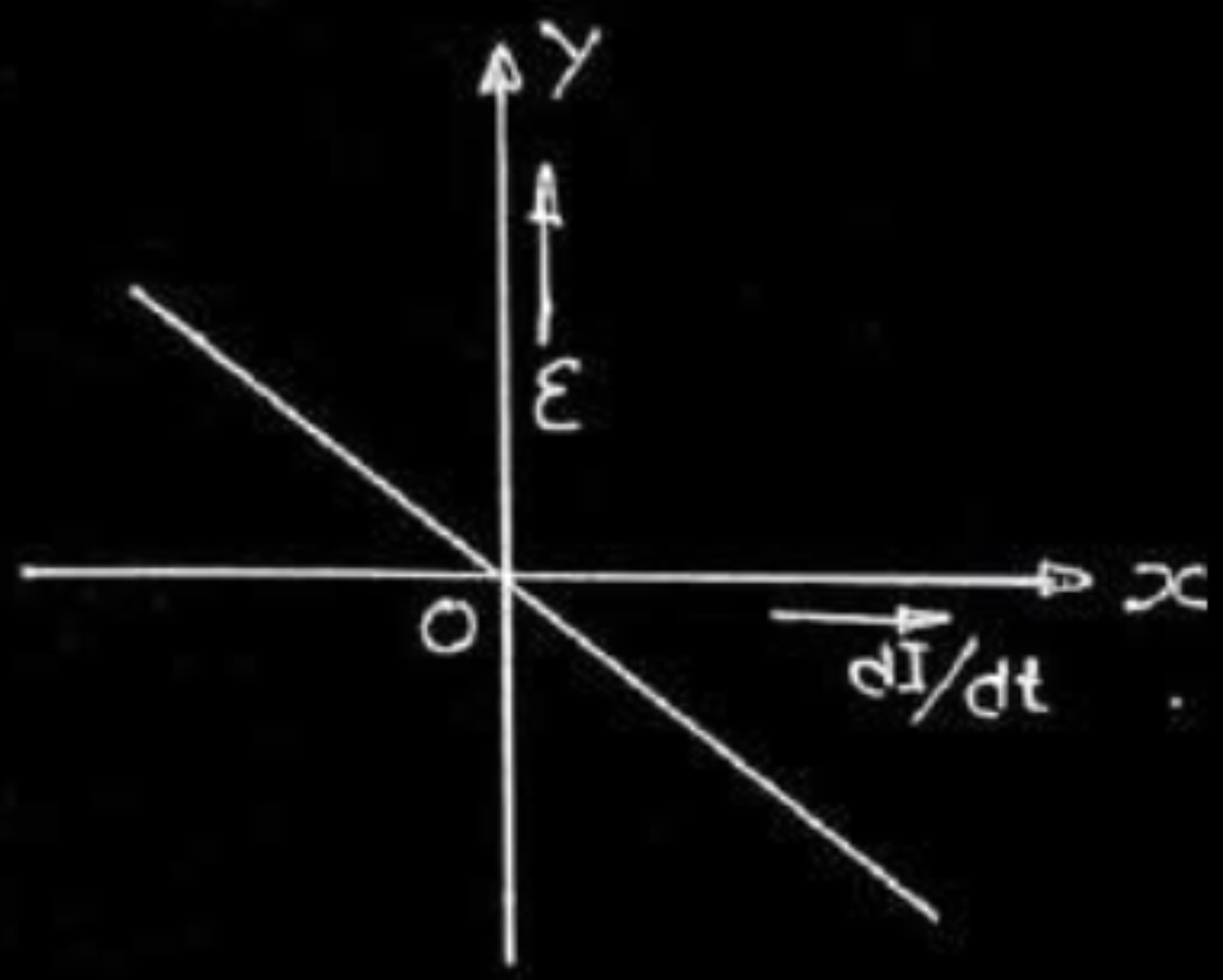
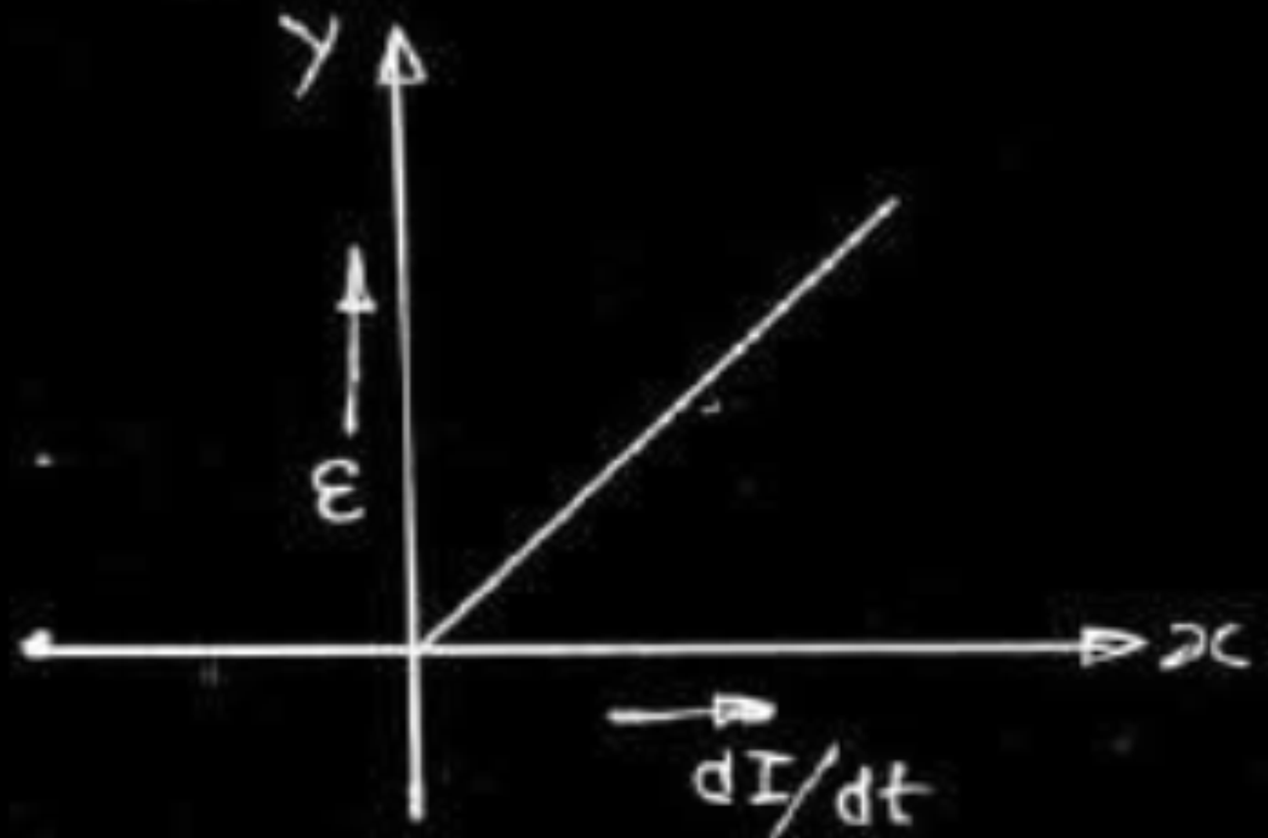


Induced e.m.f. in a coil, $\varepsilon = -L\dfrac{dI}{dt}$ ½

[Award one full mark even if the student just draws the graph without writing the expression of induced emf] ½

(Note: Award this one mark if a student draws the graph in first quadrant as shown below.)



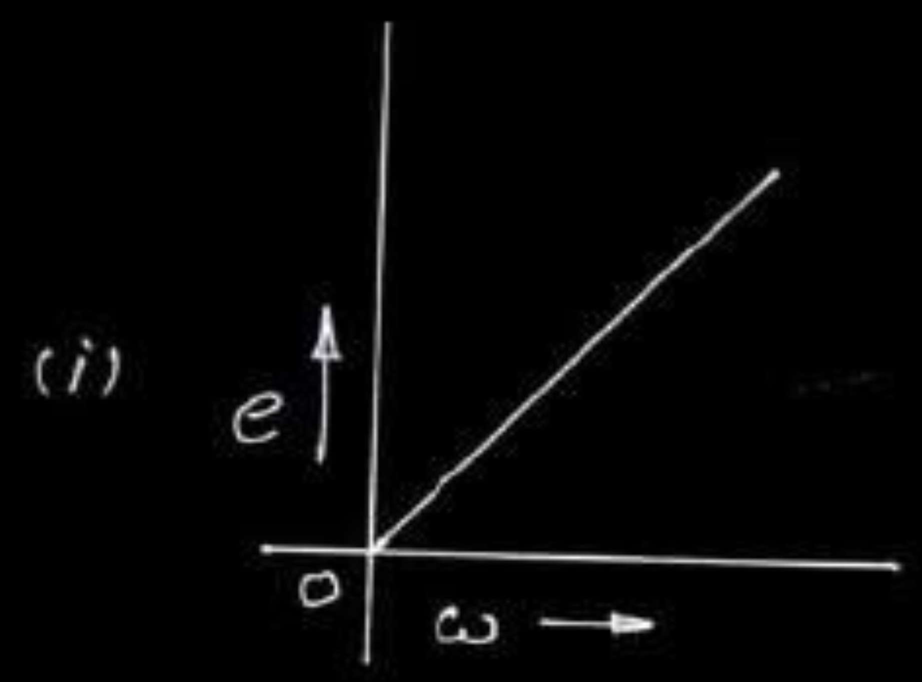
(a) A conductor of length 'l' is rotated about one of its ends at a constant angular speed 'ω' in a plane perpendicular to a uniform magnetic field B. Plot graphs to show variations of the emf induced across the ends of the conductor with (i) angular speed ω and (ii) length of the conductor l.

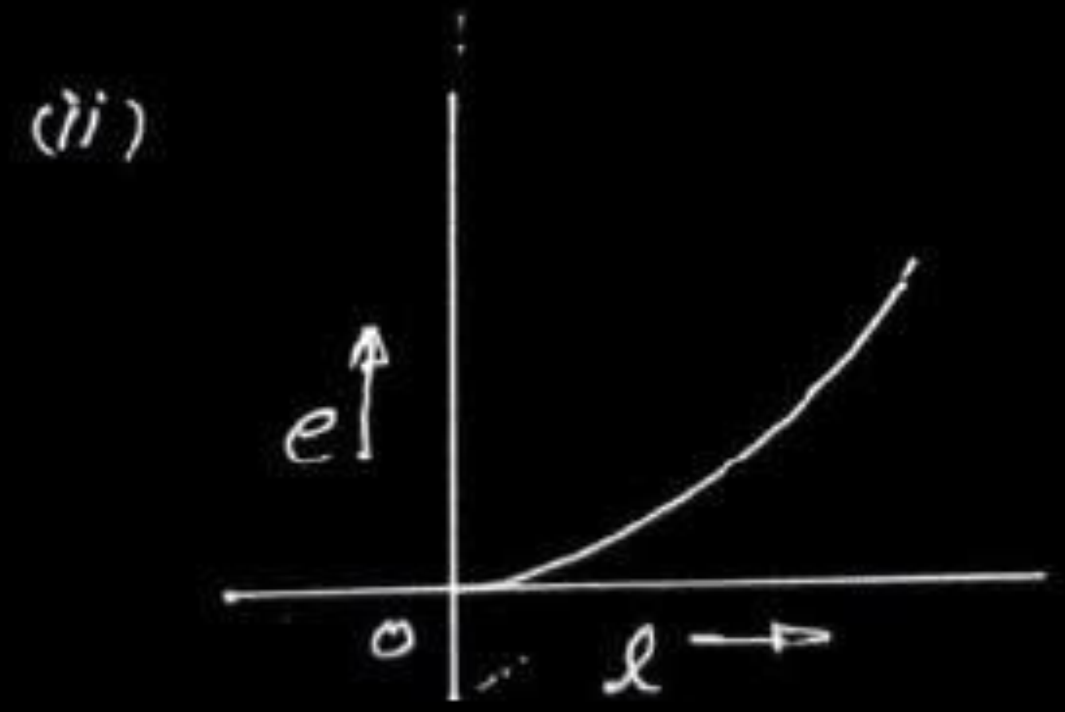
(b) Two concentric circular loops of radius 1 cm and 20 cm are placed coaxially.

(i) Find mutual inductance of the arrangement.

(ii) If the current passed through the outer loop is changed at a rate of 5 A/ms, find the emf induced in the inner loop. Assume the magnetic field on the inner loop to be uniform. **5** e.m.f. induced in the rod $= \left|-\dfrac{d\phi}{dt}\right|$ ½

$\therefore e = \dfrac{1}{2}Bl^2\omega$ ½





½+½

(i) Imagine a current I to flow through the larger coil.

Magnetic flux linked with linked with the smaller coil $= MI$

$= B_{centre} \times \pi \times 10^{-4}\,\text{Wb}$

$= \dfrac{\mu_0 I}{2\times 20\times 10^{-2}} \times \pi \times 10^{-4}\,\text{Wb}$

$\Rightarrow M = \dfrac{\mu_0 \pi}{4} \times 10^{-3}\,\text{H}$

$= \dfrac{4\pi\times 10^{-7}\pi\times 10^{-3}}{4}H$

$=9.9\times 10^{-10}H$

$=10^{-9}H$ ½

(ii)

$e.m.f.\ induced = -M\dfrac{dI}{dt}$ ½

$= -10^{-9}\times\dfrac{5}{10^{-3}}V$ ½

$= -5\times 10^{-6}V$ ½

Alternatively

$e.m.f.\ induced = -\dfrac{d\phi}{dt}$

$= -\dfrac{d}{dt}(B_c A_i) = -A_i\dfrac{dB_c}{dt}$ ½

$= -A_i\dfrac{d}{dt}\left(\dfrac{\mu_0 I}{2R}\right)$

$= -\dfrac{A_i\mu_0}{2R}\dfrac{dI}{dt}$

$= \dfrac{-\pi\times 10^{-4}\times 4\pi\times 10^{-7}\times 5}{2\times(20\times 10^{-2})\times 10^{-3}}V$ ½

$= -\dfrac{20\pi^2\times 10^{-6}}{2\times 20}V$

$= -5\times 10^{-6}V$ ½

30. (a) Differentiate between self inductance and mutual inductance.

(b) The mutual inductance of two coaxial coils is 2H. The current in one coil is changed uniformly from zero to 0.5A in 100 ms. Find the :

(i) change in magnetic flux through the other coil.

(ii) emf induced in the other coil during the change. **3**

30. (a) Differentiate between self inductance and mutual inductance.

(b) The mutual inductance of two coaxial coils is 2H. The current in one coil is changed uniformly from zero to 0.5A in 100 ms. Find the :

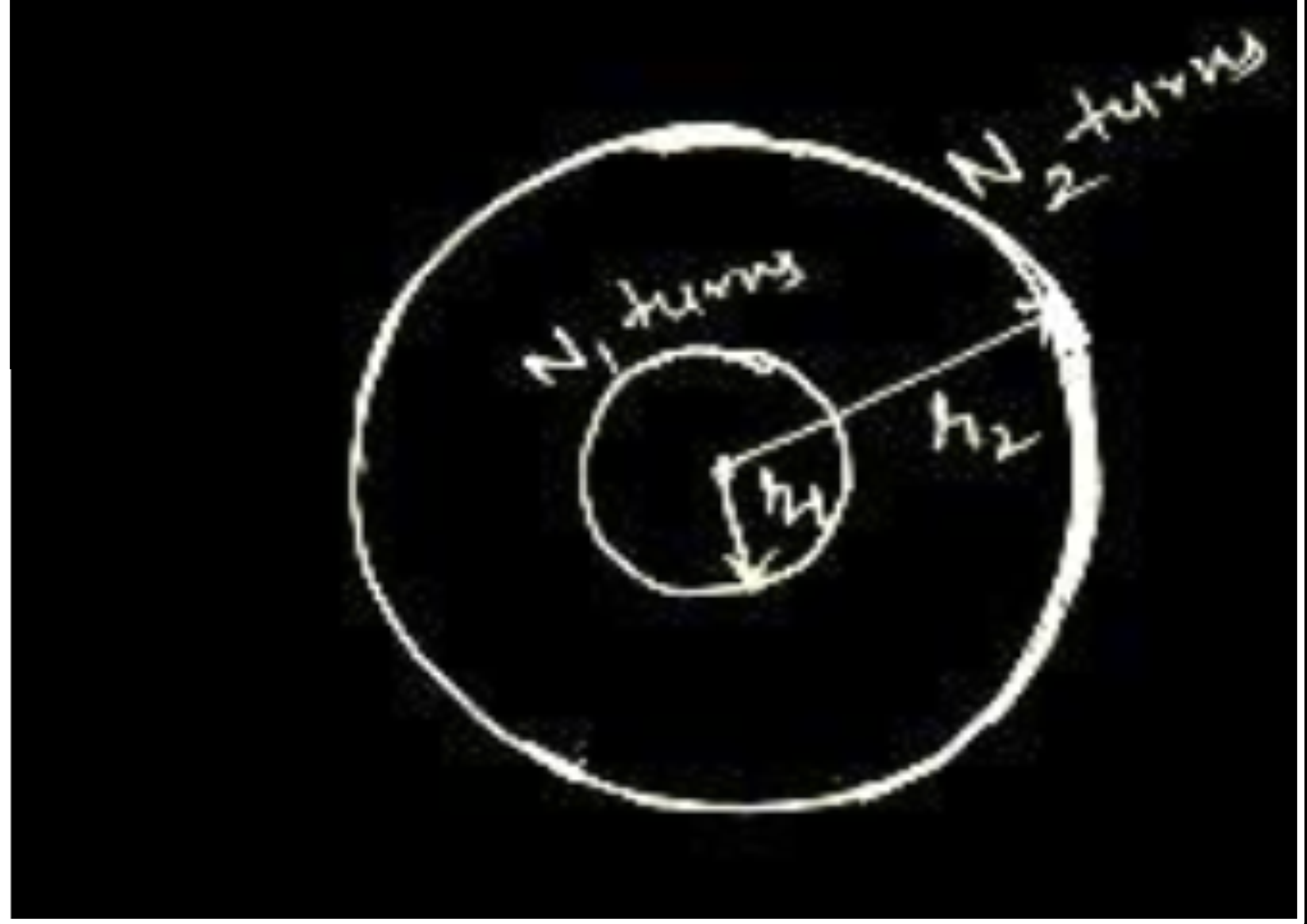
(i) change in magnetic flux through the other coil.

(ii) emf induced in the other coil during the change. **3**

(a) Self inductance is the response of the coil/ solenoid to the charge in current in the coil/ solenoid itself (or definition of self inductance)	½
Mutual inductance is the response of a coil to the charge of current in a neighbouring coil (or definition of mutual inductance)	½
Alternatively Self-inductance is the property of given coil/solenoid Mutual inductance is the property of given pair of coils /solenoids	
(b)	
(i) $\Delta\phi = M\Delta I = 2 \times 0.5 = 1Wb$	½ + ½
(ii) $e = -\frac{d\phi}{dt} = \frac{1}{100\times10^{-3}} = 10V$	½+ ½

23. Two coplanar and concentric coils 1 and 2 have respectively the number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 >> r_1$). Deduce the expression for mutual inductance of this system. *2*

The given system has the shape shown here
Let a current I flow through the larger coil.



Magnetic field, due to the current at the centre of coil is

$$B_C = \frac{\mu_o I N_2}{2r_2}$$ ½

We can consider this to be the value of the magnetic field over the whole area of the smaller coil(as $r_1 << r_2$)

∴Magnetic flux through the smaller coil

$$= B_c(\pi r_1^2)N_1 = \frac{\mu_o I N_2}{2r_2}\pi r_1^2 N_1$$

$$= \left(\frac{\mu_o \pi r_1^2}{2r_2}N_1 N_2\right)I$$ ½

But Magnetic flux=MI

Where M= mutual Inductance of the system

$$\therefore MI = \frac{\mu_o \pi r_1^2 N_1 N_2}{2r_2}I$$

$$M = \frac{\mu_o \pi r_1^2 N_1 N_2}{2r_2}$$ ½

18. Draw the graph showing variation of the value of the induced emf as a function of rate of change of current flowing through an ideal inductor. *1*

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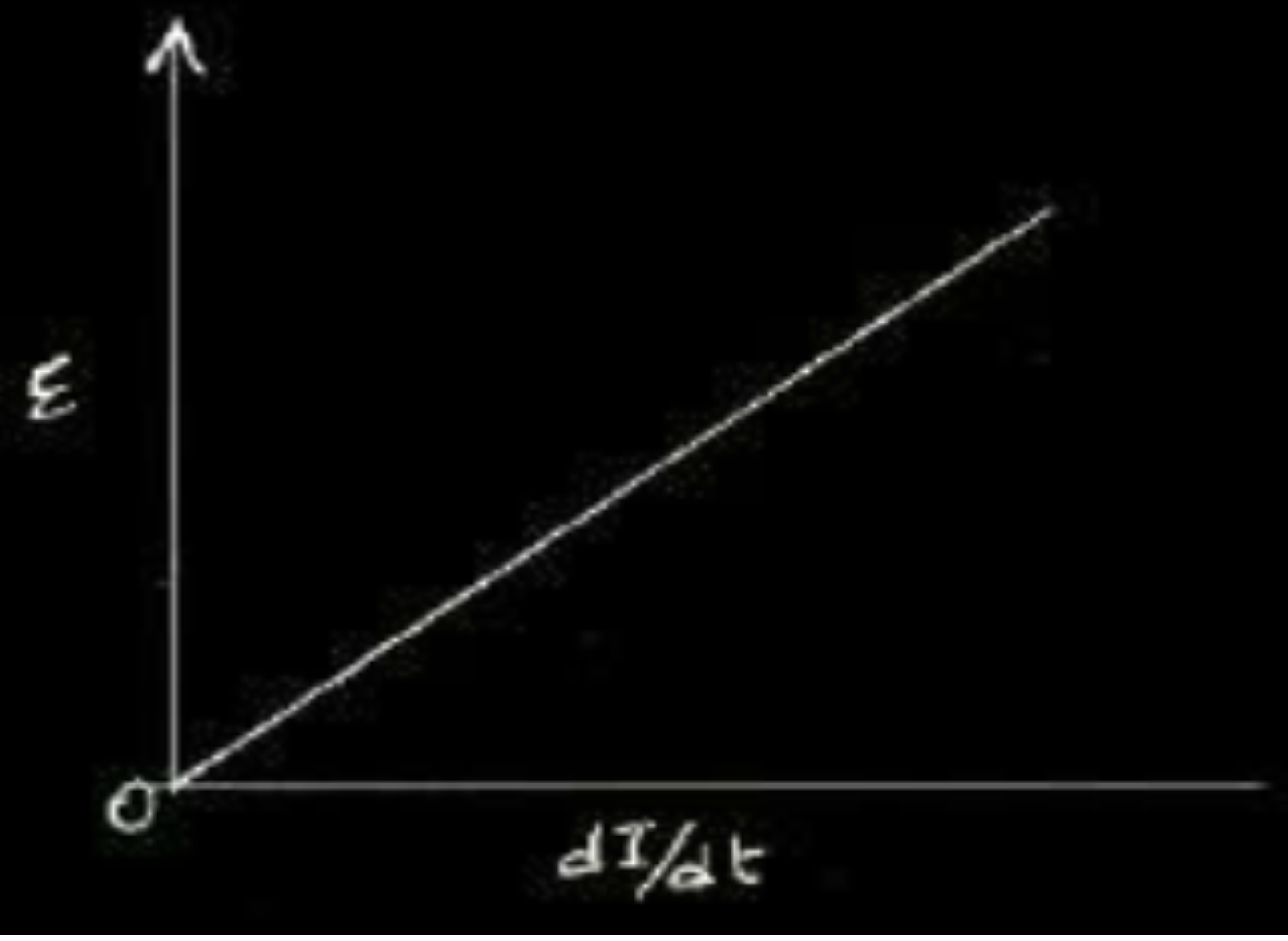
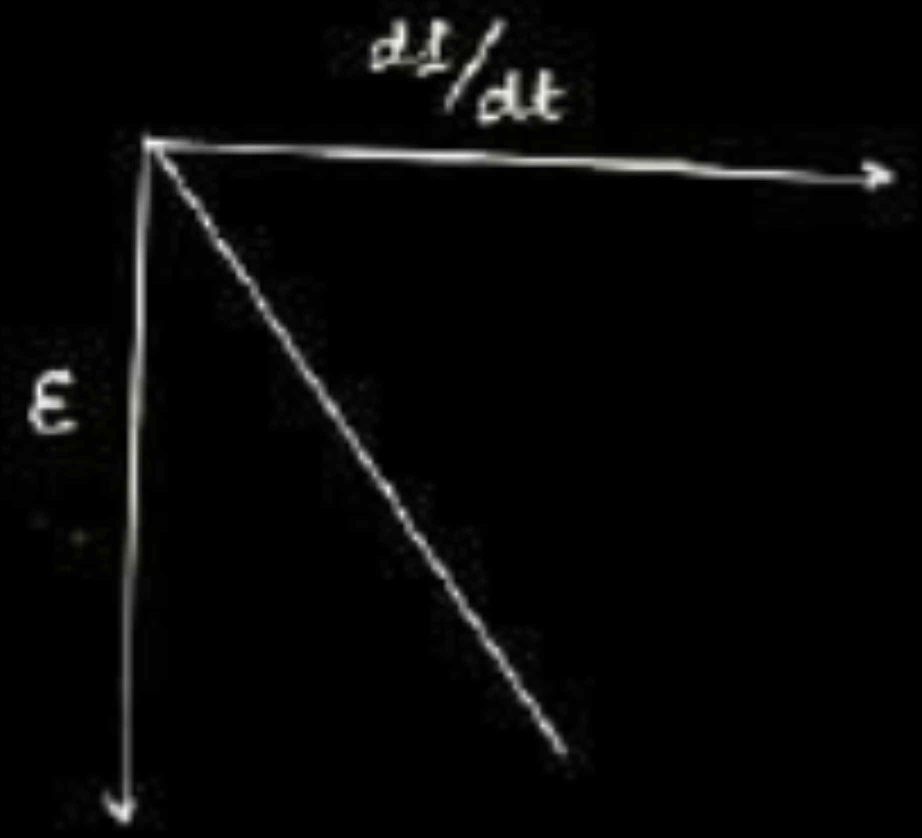
 Alternatively 	1

13. The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become ___________ times. *1*

13. The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become ___________ times. *1*

Four times

(a) A rectangular coil rotates in a uniform magnetic field. Obtain an expression for induced emf and current at any instant. Also find their peak values. Show the variation of induced emf versus angle of rotation (ωt) on a graph.

(b) An iron bar falling through the hollow region of a thick cylindrical shell made of copper experiences a retarding force. What can you conclude about the nature of the iron bar ? Explain.

(a)

$$\phi = N\vec{B}.\vec{A}$$ ½

$$\phi = NBA\cos\omega t$$ ½

$$\varepsilon = \frac{-d\phi}{dt}$$ ½

$$\varepsilon = NBA\omega\sin\omega t$$ ½

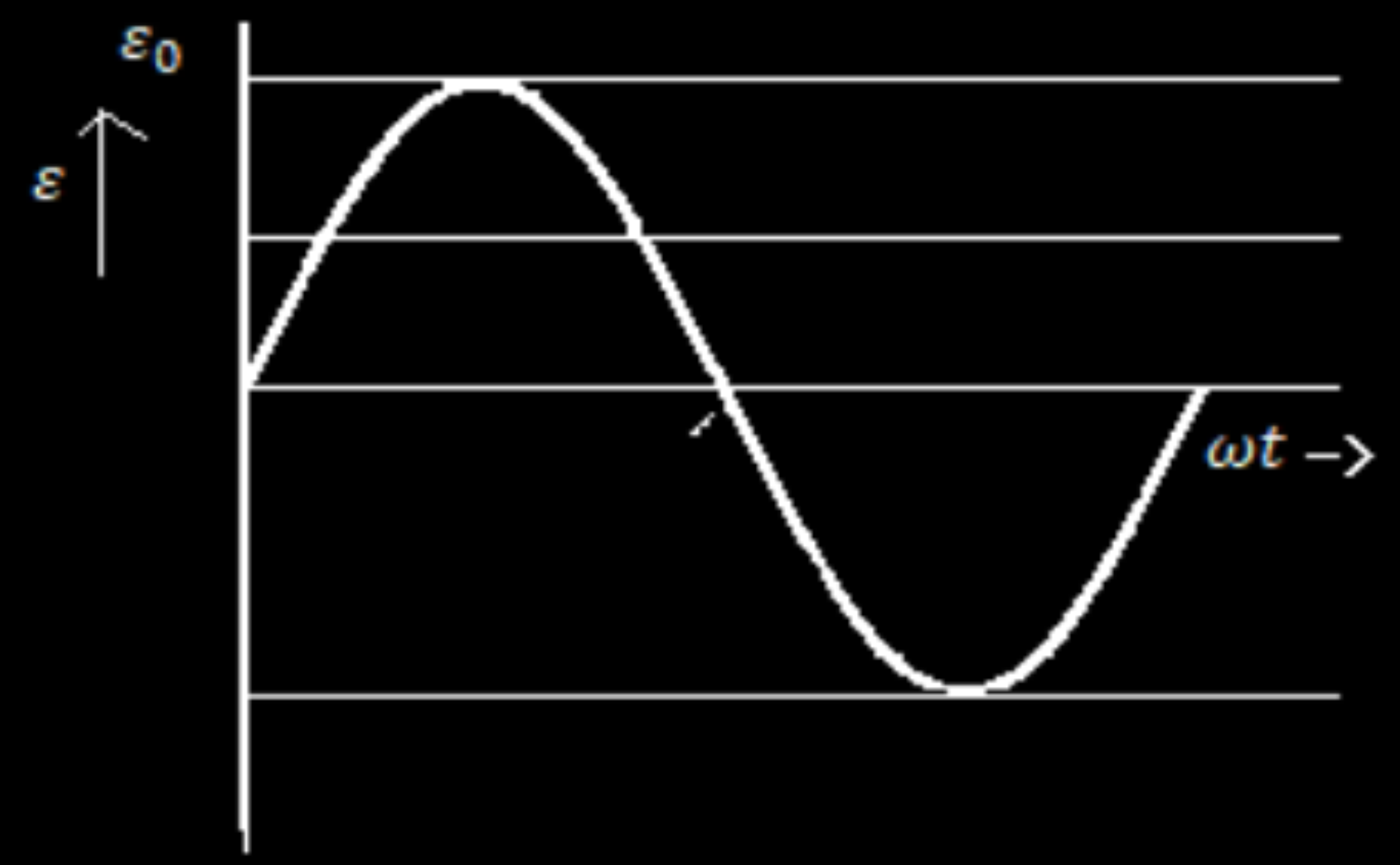
$$here \quad \varepsilon_0 = NBA\omega$$ ½

$$i = \frac{\varepsilon}{R} = \frac{NBA\omega}{R}\sin\omega t$$ ½

$$i_0 = \frac{NBA\omega}{R}$$ ½

(b)Bar is magnetic

<u>Reason:</u> Lenz's law/(Induced emf/current opposes its cause) ½



(a) Draw a schematic diagram of an ac generator. Explain its working and obtain the expression for the instantaneous value of the emf in terms of the magnetic field B, number of turns N of the coil of area A rotating with angular frequency ω. Show how an alternating emf is generated by a loop of wire rotating in a magnetic field.

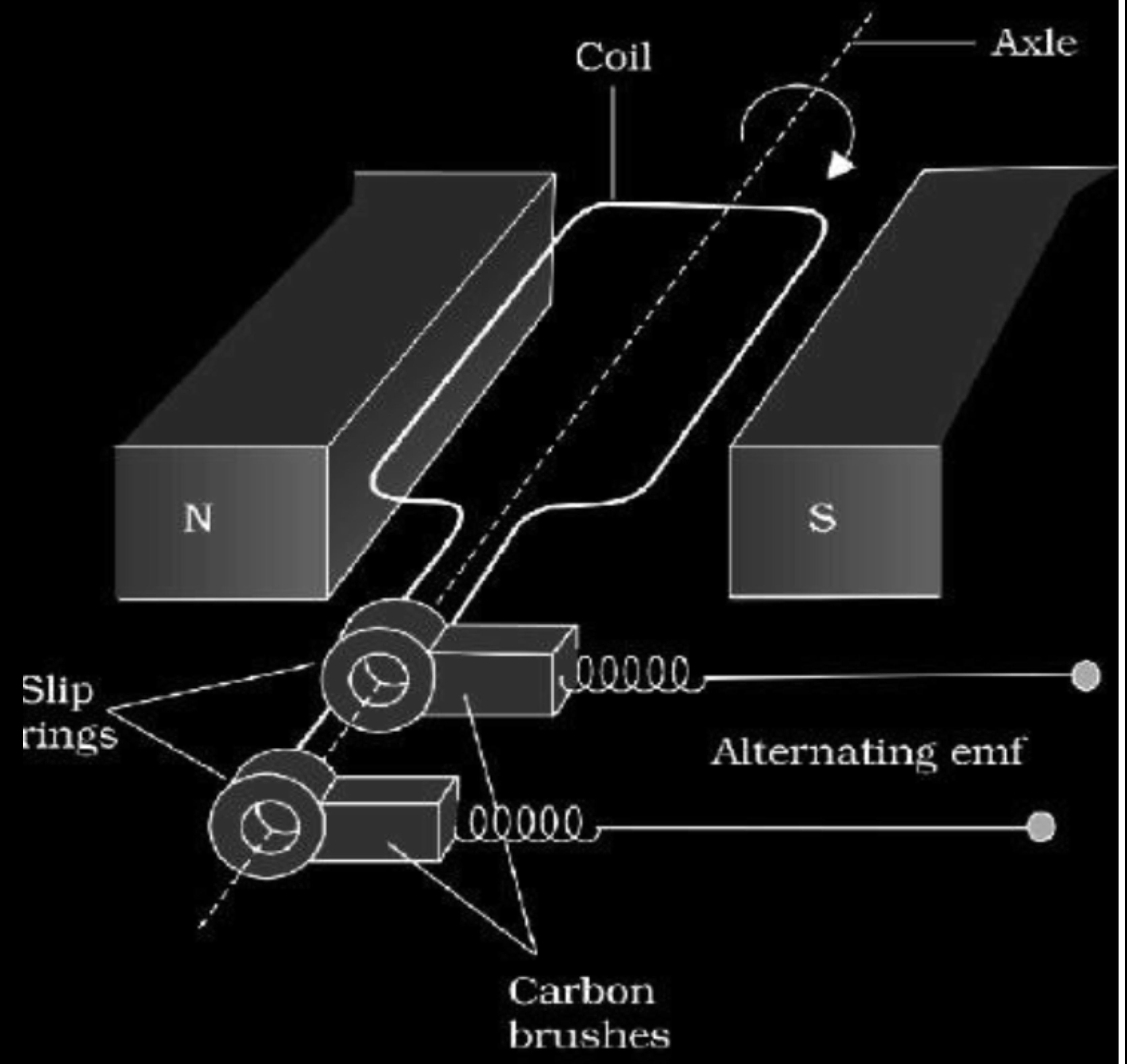
(b) A circular coil of radius 10 cm and 20 turns is rotated about its vertical diameter with angular speed of 50 rad s^{-1} in a uniform horizontal magnetic field of $3{\cdot}0 \times 10^{-2}$ T.

(i) Calculate the maximum and average emf induced in the coil.

(ii) If the coil forms a closed loop of resistance 10 Ω, calculate the maximum current in the coil and the average power loss due to Joule heating.

Working of AC Generator -

Whenever coil placed in uniform magnetic field is rotated, flux linked with it changes, and an emf induces in the coil. The ends of the coil are connected to an external circuit by means of slip rings and brushes. ½



Flux linked with the coil of Area a, placed in uniform magnetic field 'B'

$$\phi_B = BA\cos\theta$$

or $\phi_B = BA\cos\omega t$ .... (eq^{n}1)

$\therefore$ From Faraday's law

e.m.f induced in the coil

$$\varepsilon = -N\frac{d\phi_B}{dt}$$ ½

$$-NBA\frac{d}{dt}\cos\omega t$$

$$\varepsilon = NBA\omega\sin\omega t$$

or $\varepsilon = \varepsilon_0\sin\omega t$ where $\varepsilon_0 = NBA\omega$ ½

b) i) $r = 10$ cm, $N = 20$ turns, $\omega = 50$ rad s^{-1}

$B = 3.0 \times 10^{-2}$ T

$\varepsilon_0 = NBA\omega$

$= 20 \times 3 \times 10^{-2} \times \pi \left(10 \times 10^{-2}\right)^2 \times 50$

$= 0.942$ volt ½

$\varepsilon_{AV} = 0$, over a cycle

½

ii) $i_0 = \dfrac{e_0}{R} = \dfrac{0.942}{10}$

$= 0.094$ A ½

$P = \dfrac{1}{2}\ \varepsilon_0 \times I_0$

½

$= \dfrac{1}{2} \times 0.942 \times 0.094$

$= 0.045$ watt. ½

A 0·5 m long solenoid of 10 turns/cm has area of cross-section 1 cm^2. Calculate the voltage induced across its ends if the current in the solenoid is changed from 1 A to 2 A in 0·1 s.

A 0·5 m long solenoid of 10 turns/cm has area of cross-section 1 cm^2. Calculate the voltage induced across its ends if the current in the solenoid is changed from 1 A to 2 A in 0·1 s.

Induced voltage	
$\lvert V \rvert = L\dfrac{dI}{dt}$	½
$\therefore \lvert V \rvert = \mu_0 n^2 la \dfrac{dI}{dt}$	½
$= 4\pi\times10^{-7}\times\left(\dfrac{10}{10^{-2}}\right)^2 \times 0.5 \times 1 \times 10^{-4} \times \dfrac{(2-1)}{0.1}$	½
$= 6.28\times10^{-4}$ V or 0.628mV	½

(a) Derive an expression for the induced emf developed when a coil of N turns, and area of cross-section A, is rotated at a constant angular speed ω in a uniform magnetic field B.

(b) A wheel with 100 metallic spokes each 0·5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of the Earth's magnetic field. If the resultant magnetic field at that place is 4×10^{-4} T and the angle of dip at the place is 30°, find the emf induced between the axle and the rim of the wheel.

Flux linked with the coil at any instant of time is:	
$\phi = NBA\cos\omega t$	½
$\dfrac{d\phi}{dt} = NBA\omega(-\sin\omega t)\,S$	
$\varepsilon = -\dfrac{d\phi}{dt}$	1
$\varepsilon = NBA\omega\sin\omega t$	½
$\varepsilon = \varepsilon_0 \sin\omega t \quad (\text{Here } \varepsilon_0 = NBA\omega)$	
(b) $l = 0.5m,\ \nu = 120rpm = 2rps$	½
$\omega = 2\pi\nu = 4\pi\ rad/s, \quad B = 4\times10^{-4}T, \quad \delta = 30^0$	
$B_H = 4\times10^{-4}\times\dfrac{\sqrt{3}}{2}$	
$B_H = 2\sqrt{3}\times10^{-4}T$	½
$\varepsilon = \dfrac{1}{2}B\omega l^2$	1
$\varepsilon = \dfrac{1}{2}\times2\sqrt{3}\times10^{-4}\times4\pi\times(0.5)^2$	
$\varepsilon = 5.4\times10^{-4}\,volt$	1

(a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.

(a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.

(a) When external source supplies current to the inductor, e.m.f. is induced in it due to self induction. So the external supply has to do work to establish current. The amount of work done is: $dw = \lvert\varepsilon\rvert Idt \qquad \because \varepsilon = L\dfrac{dI}{dt}$ $dw = LIdt$	½
$W = \dfrac{1}{2}LI^2$ Energy density $= \dfrac{Energy}{Volume}$ $u = (1/2LI^2)/volume$	1 ½

(a) Define mutual inductance and write its S.I. unit.

(a) Define mutual inductance and write its S.I. unit.

(a) Mutual inductance equals the magnetic flux associated with a coil when unit current flows in its neighbouring coil. Alternatively: Mutual inductance equals the induced emf in a coil when the rate of change of current in its neighbouring coil is one ampere/ second.	1
S.I unit : henry (H) or weber/ampere (or any other correct SI unit)	½

(a) Define the term 'self-inductance' and write its S.I. unit.

(b) Obtain the expression for the mutual inductance of two long co-axial solenoids S_1 and S_2 wound one over the other, each of length L and radii r_1 and r_2 and n_1 and n_2 number of turns per unit length, when a current I is set up in the outer solenoid S_2.

a) Definition of self inductance and its SI unit 1 + ½ b) Derivation of expression for mutual inductance 1 ½	
Self inductance of a coil equals, the magnitude of the magnetic flux, linked with it, when a unit current flows through it. Alternatively Self inductance, of a coil, equals the magnitude of the emf induced in it, when the current in the coil, is changing at a unit rate.	1
SI unit : henry / (weber/ampere) / (ohm second.)	½

(a) Draw a labelled diagram of AC generator. Derive the expression for the instantaneous value of the emf induced in the coil.

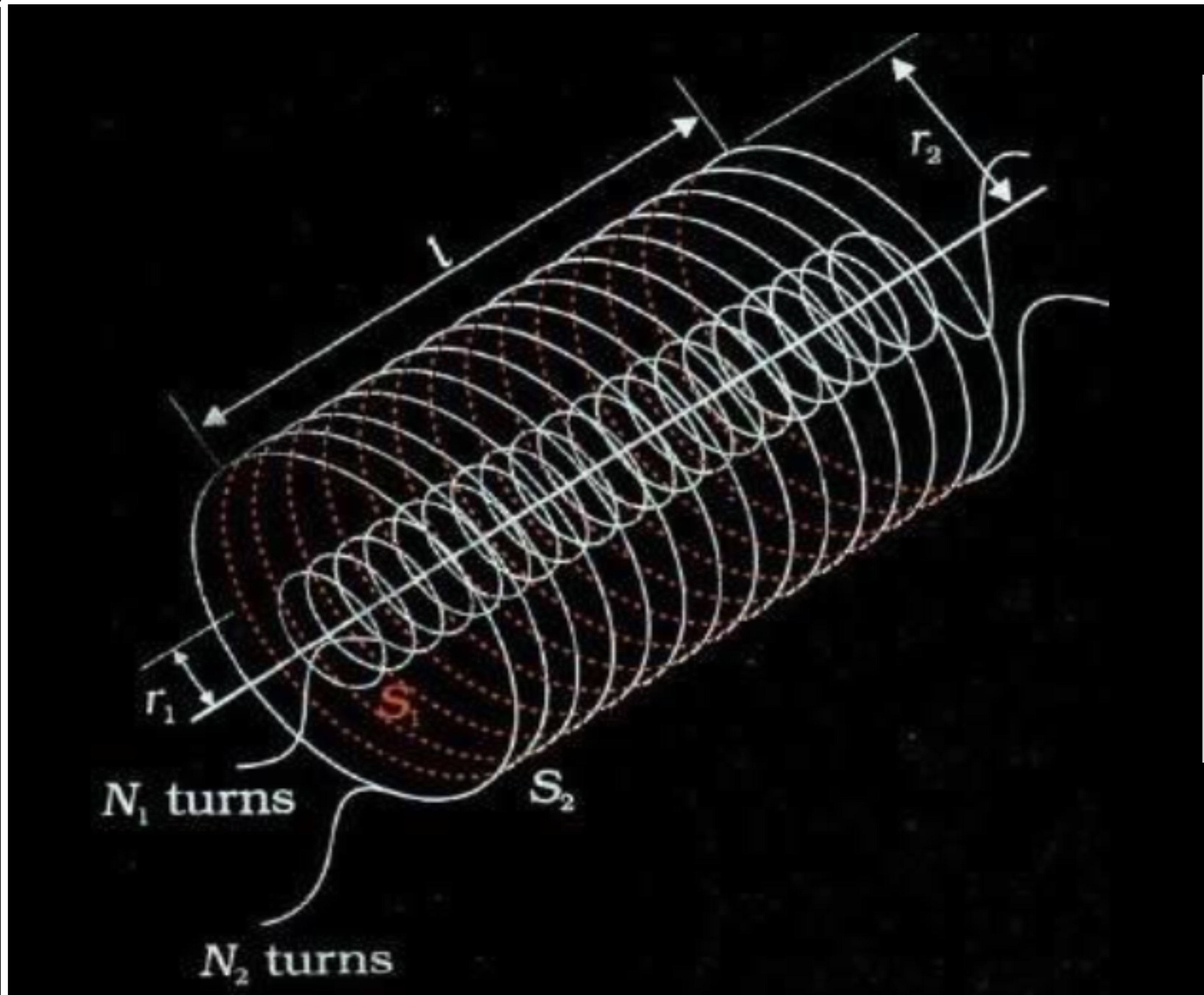
(b) A circular coil of cross-sectional area 200 cm^2 and 20 turns is rotated about the vertical diameter with angular speed of 50 rad s^{-1} in a uniform magnetic field of magnitude 3.0×10^{-2}T. Calculate the maximum value of the current in the coil.

[Deduct ½ mark, If diagram is not labeled] When the coil is rotated with constant angular speed ω, the angle θ between the magnetic field and area vector of the coil, at instant t, is given by θ $= \omega$ t, Therefore, magnets flux, (ϕ_B), at this instant, is $= BA \cos \omega t$	½
∴Induced emf $e = -N$——	½
$e = NBA\, \omega \sin \omega t$ $e = e_o \sin \omega t$ where e_o= NBA	½
b) Maximum value of emf	½
$= NBA\, \omega$	½
= 20 x 200 x x 3 x x 50V = 600 mV	½
Maximum induced current = — ——mA	½
[Note 1: It the student calculates the value of the maximum induced emf and says that " since R is not given, the value of maximum induced current cannot be calculated", the ½ mark, for the last part, of the question, can be given.] [Note 2: The direction of magnetic field has not been given. If the student takes this direction along the axis of rotation and hence obtains the value of induced emf and, therefore, maximum current, as zero, award full marks for this part.]	

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

(i) Mutual inductance is numerically equal to the induced emf in the secondary coil when the current in the primary coil changes by unity. <u>Alternatively:</u> Mutual inductance is numerically equal to the magnetic flux linked with one coil/secondary coil when unit current flows through the other coil /primary coil.	**1**
Let a current, i_2, flow in the secondary coil $$\therefore B_2 = \frac{\mu_0 N_2 i_2}{l}$$	½
$\therefore$ Flux linked with the primary coil $$= N_1 A_1 B_2 = \frac{\mu_0 N_2 N_1 A_1 i_2}{l} = M_{12} i_2$$	½
Hence, $M_{12} = \frac{\mu_0 N_2 N_1 A_2}{l} = \mu_0 n_2 n_1 A_1 l \left(n_1 = \frac{N_1}{l}; n_2 = \frac{N_2}{l}\right)$	½



½

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

(i)	Self inductance, of a coil, is numerically equal to the emf induced in that coil when the current in it changes at a unit rate. (Alternatively: The self inductance of a coil equals the flux linked with it when a unit current flows through it.)	**1**
(ii)	The work done against back /induced emf is stored as magnetic potential energy. The rate of work done, when a current i is passing through the coil, is $\frac{dW}{dt} = \lvert\varepsilon\rvert i = \left(L\frac{di}{dt}\right) i$ $\therefore W = \int dW = \int_0^I Lidi$ $= \frac{1}{2}Li^2$	½ ½ ½ ½

CBSE 2016

(i) Define mutual inductance.

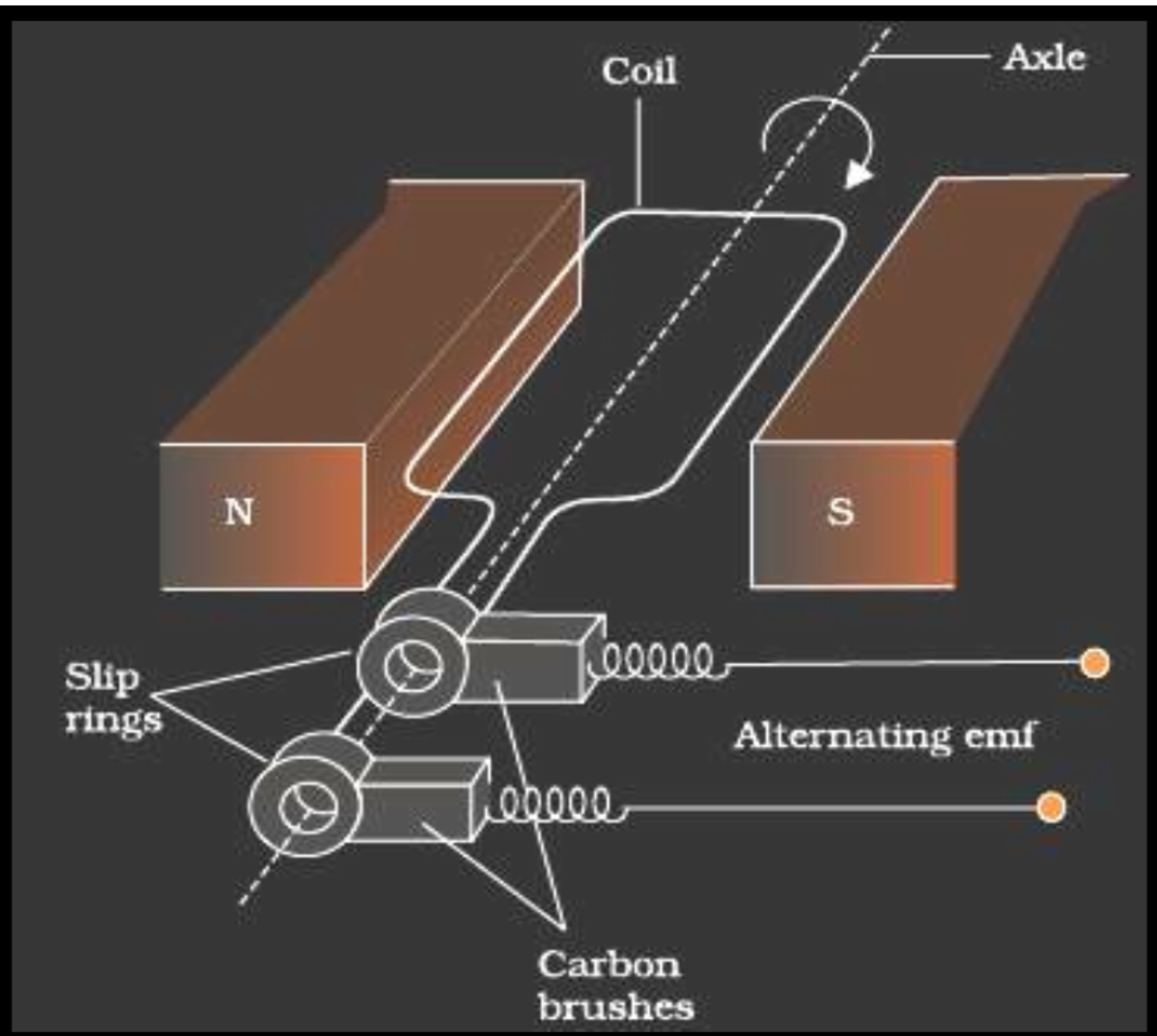
(ii) A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of flux linkage with the other coil ?

(i) Magnetic flux, linked with the secondary coil due to the unit current flowing in the primary coil, $\phi_2 = MI_1$ [Alternatively Induced emf associated with the secondary coil, for a unit rate of change of current in the primary coil. $e_2 = -M\frac{dI_1}{dt}$] [Also accept the Definition of Mutual Induction, as per the Hindi translation of the question] [i.e. the phenomenon of production of induced emf in one coil due to change in current in neighbouring coil]	1
(ii) Change of flux linkage $d\phi = M\ dI$	1
$= 1.5 \times (20\text{-}0)$W	½
= 30 weber	½

(a) Explain the meaning of the term mutual inductance. Consider two concentric circular coils, one of radius r_1 and the other of radius r_2 ($r_1 < r_2$) placed coaxially with centres coinciding with each other. Obtain the expression for the mutual inductance of the arrangement.

(b) A rectangular coil of area A, having number of turns N is rotated at ' f ' revolutions per second in a uniform magnetic field B, the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2\pi f NBA$.

a. Mutual Inductance is the property of a pair of coils due to which an emf induced in one of the coils due to the change in the current in the other coil.	½
Mathematically $e_2 = -\frac{Mdi_1}{dt}$ $\therefore M = -\frac{e_2}{di_1/dt}$	½
Let a current I_2 flow through the outer circular coil. Then $B_2 = \mu I_2/2r_2$ $\therefore \emptyset_1 = \pi r^2 B_2 = \frac{\mu\pi r_1^2}{2r_2} I_2 = M_{12} I_2$ Thus $M_{12} = \frac{\mu\pi r_1^2}{2r_2} = M_{21}$	

 	½ ½
Flux at any time 't'. $\emptyset_B = BAcos\theta = BAcos\omega t$ From Farday's Law, induced emf $e = -N\frac{d\emptyset_B}{dt} = NBA\frac{d}{dt}(\cos\omega t)$ Thus the instantaneous value of emf is E= NBAωsinωt For maximum value of emf sin$\omega t = \pm 1$ i.e, $e_0 = NBAw = 2\pi fNBA$	½ ½